

Electronics

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Recommended Books

Electronic Devices and Circuit Theory by Robert L. Boylestad and Louis Nashelsky

Microelectronic Circuits by Adel S. Sedra and Kenneth C. Smith

Electronic Devices by Thomas L. Floyd

Marks Distribution

Mid – 20 (Average of 2 exams)

Attendance – 10

Evaluation – 10 (Class Performance)

Final – 60 - (Proper order, proper English)

Electronics

Electronics?

- A scientific and engineering discipline
- Study that applies the principles of physics to design, create, and operate devices that manipulate electrons and other electrically charged particles.
- Focuses on using active devices such as transistors, diodes, and integrated circuits to control and amplify the flow of electric current
- To convert it from one form to another, such as from alternating current (AC) to direct current (DC) or from analog to digital.
- Electronics also encompasses the fields of microelectronics, nanoelectronics, optoelectronics, and quantum electronics, which deal with the fabrication and application of electronic devices at microscopic, nanoscopic, optical, and quantum scales.

-Wikipedia

Material Types

Conductor: Supports a generous flow of charge

- Copper, Aluminium etc.

Insulator: Very low level of conductivity

- Mica, rubber, glass etc.

Semiconductor: Conductivity level some-where between the conductor and insulator.

- Silicon, germanium, gallium arsenide etc.

Conductivity and Resistivity

Conductivity: Support to the flow of charge.

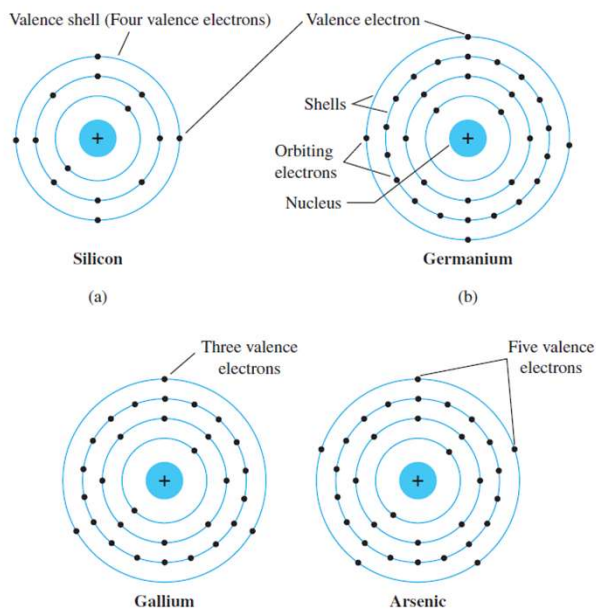
Resistivity: Inversely related to the conductivity which is the resistance to the flow of charge.

$$\rho = \frac{RA}{l} \text{ } \Omega\text{-cm}$$

TABLE 1.1 Typical Resistivity Values

<i>Conductor</i>	<i>Semiconductor</i>	<i>Insulator</i>
$\rho \cong 10^{-6} \text{ } \Omega\text{-cm}$ (copper)	$\rho \cong 50 \text{ } \Omega\text{-cm}$ (germanium) $\rho \cong 50 \times 10^3 \text{ } \Omega\text{-cm}$ (silicon)	$\rho \cong 10^{12} \text{ } \Omega\text{-cm}$ (mica)

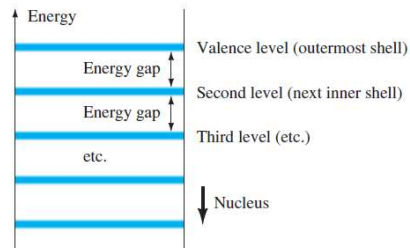
Atomic structure of Intrinsic Semiconductor



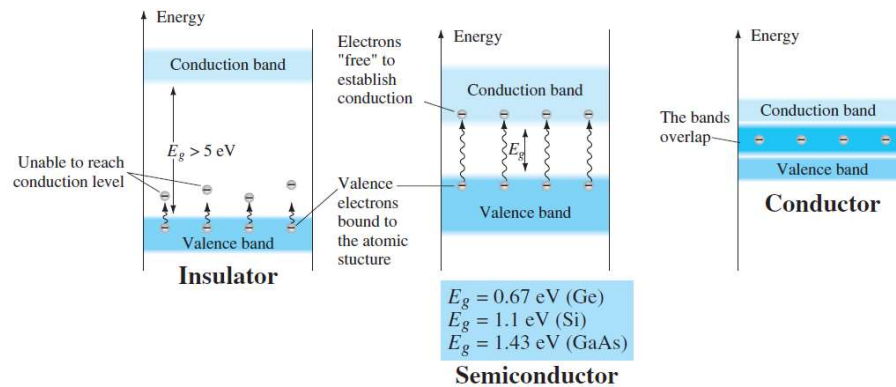
- Single-crystal semiconductors: germanium (Ge) and silicon (Si)
- Compound semiconductors : Gallium Arsenide (GaAs), Cadmium Sulfide (CdS), etc
(constructed of two or more semiconductor materials of different atomic structures.)
- Ge, Si, and GaAs are most frequently used in the construction of electronic devices.
- Intrinsic carriers: Free carriers due to the natural causes only.

ENERGY LEVELS

SEMICONDUCTOR
DIODES



(a)

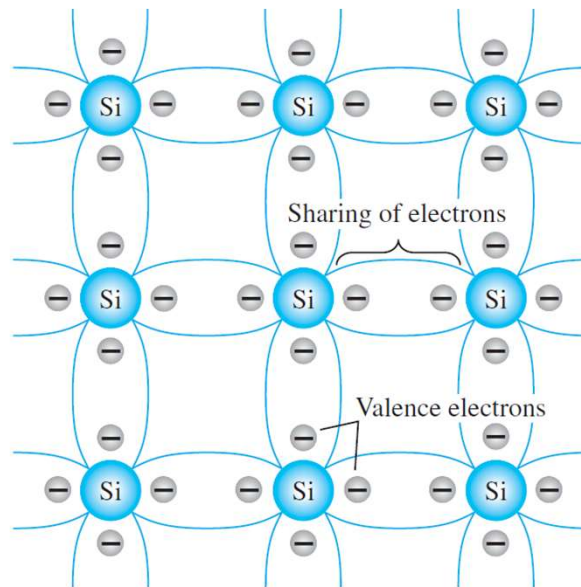


The farther an electron is from the nucleus, the higher is the energy state, and any electron that has left its parent atom has a higher energy state than any electron in the atomic structure.

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

Atomic structure of Intrinsic Semiconductor

- During the intrinsic state the Si is still a poor conductor.



Extrinsic Semiconductor Materials

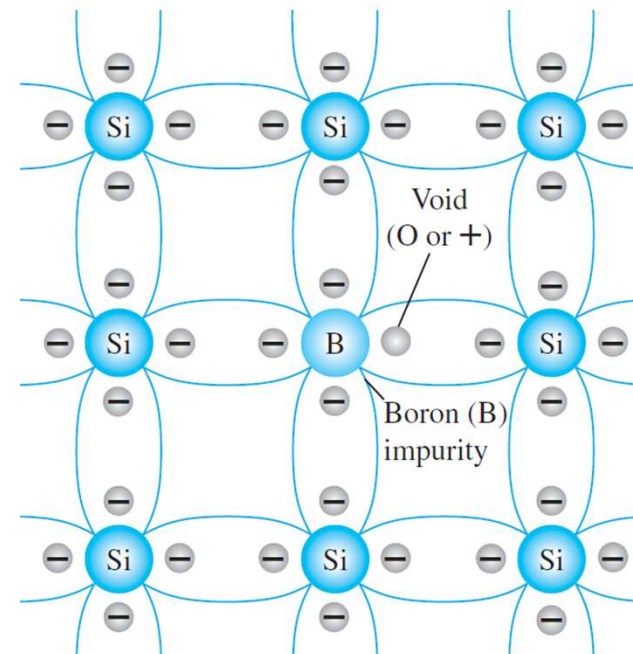
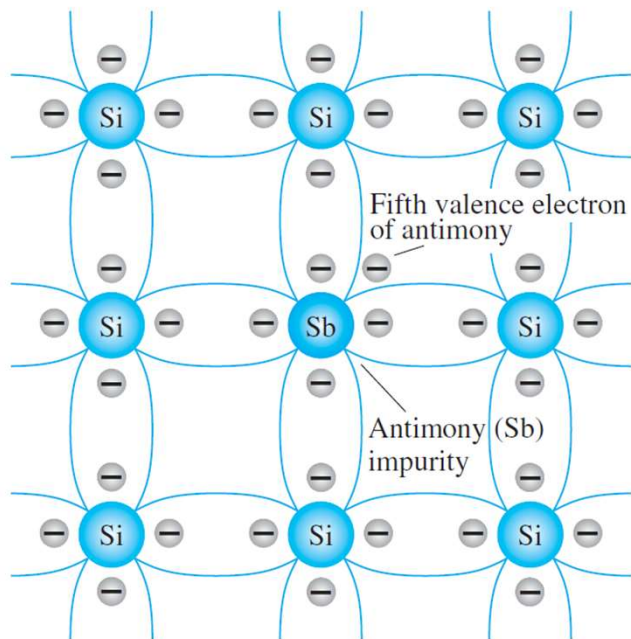
Doping: The process of adding impurities to the intrinsic semiconductor materials.

Extrinsic semiconductor material: Intrinsic + Doping process.

n-Type extrinsic semiconductor materials: Impurities with 5 valence electrons such as antimony, arsenic and phosphorus.

p-Type extrinsic semiconductor materials: Impurities with 3 valence electrons such as boron, gallium and indium.

n-Type and p-Type Semiconductor Materials

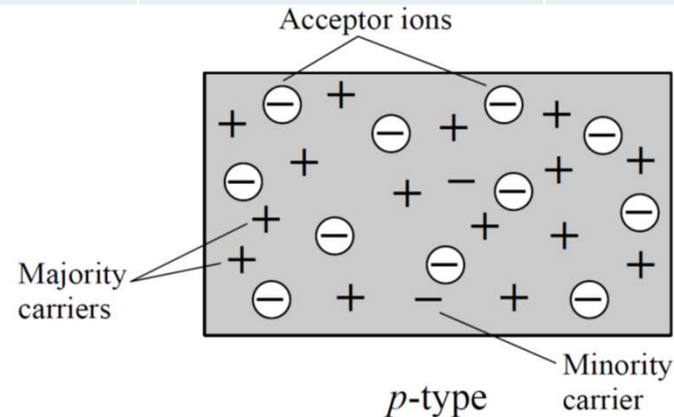
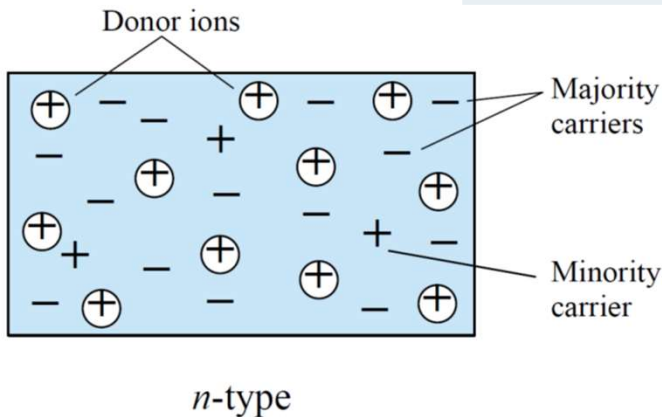


n-Type and p-Type Semiconductor Materials Cont'd

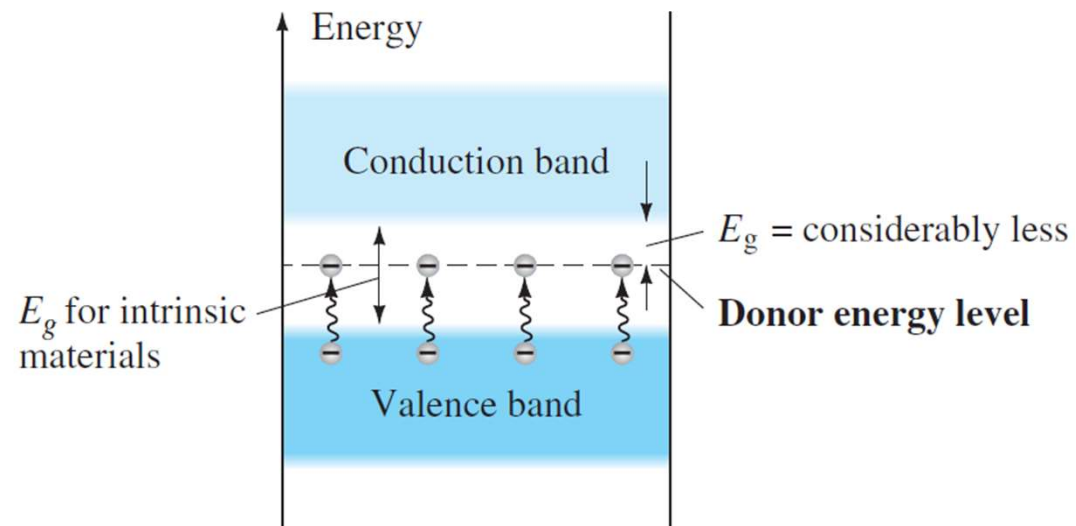
Impurities with 5 valence electrons are called donor atoms.

Impurities with 3 valence electrons are called acceptor atoms.

Types of Materials	Majority Carriers	Minority Carriers
n-Type	Electrons	Holes
p-Type	Holes	Electrons



n-Type Semiconductor Materials Cont'd



Majority and Minority Carriers

- In an n-type material the electron is called the majority carrier and the hole the minority carrier.
- In a p-type material the hole is the majority carrier and the electron is the minority carrier.

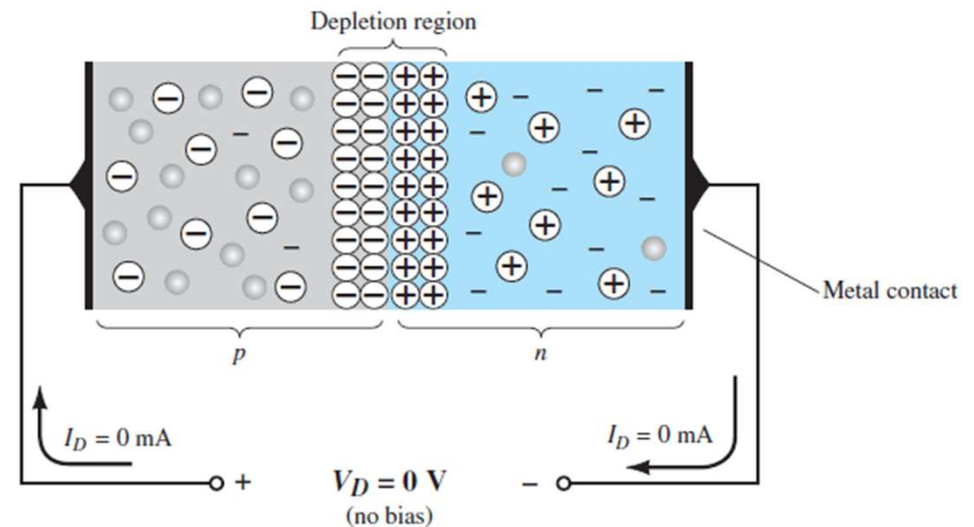
Semiconductor Diode

Simply joining an n -type and a p -type material together.

A two terminal device !

Depletion region : free carriers have been absorbed.

Contact potential/ barrier potential/ space charge potential : The electric field formed in the depletion region acts as a barrier. External energy must be applied to get the electrons to move across the barrier of the electric field. The potential difference required to move the electrons through the electric field is called the barrier potential.

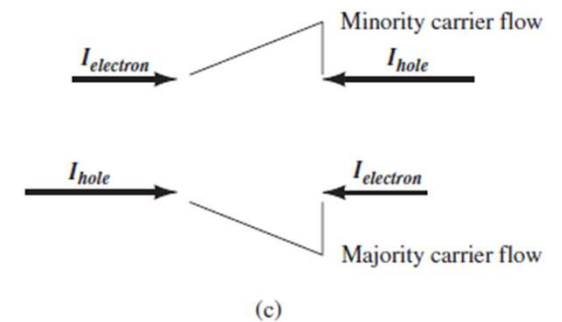
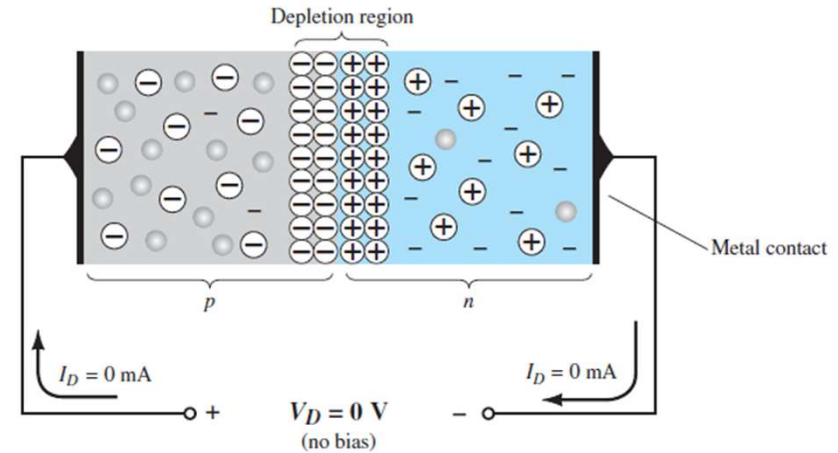
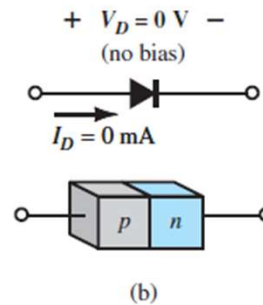
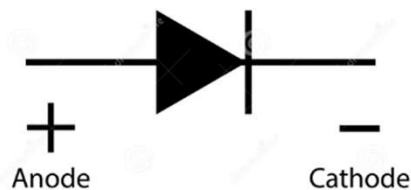


Semiconductor Diode

Bias : The term *bias* refers to the application of an external voltage across the two terminals of the device to extract a response.

No bias ($V_D = 0$)

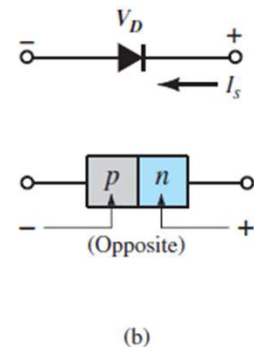
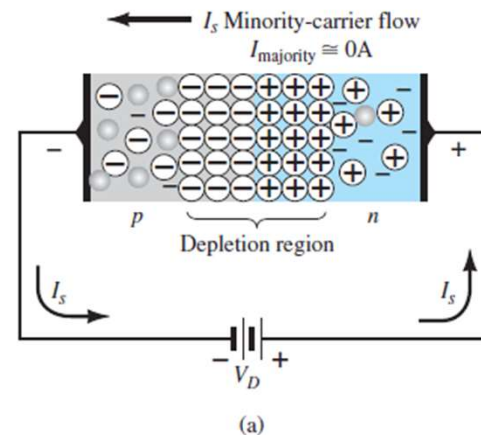
- Net flow in either direction is zero.



Semiconductor Diode Cont'd

Reverse bias ($V_D < 0$)

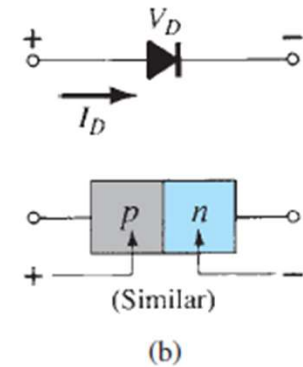
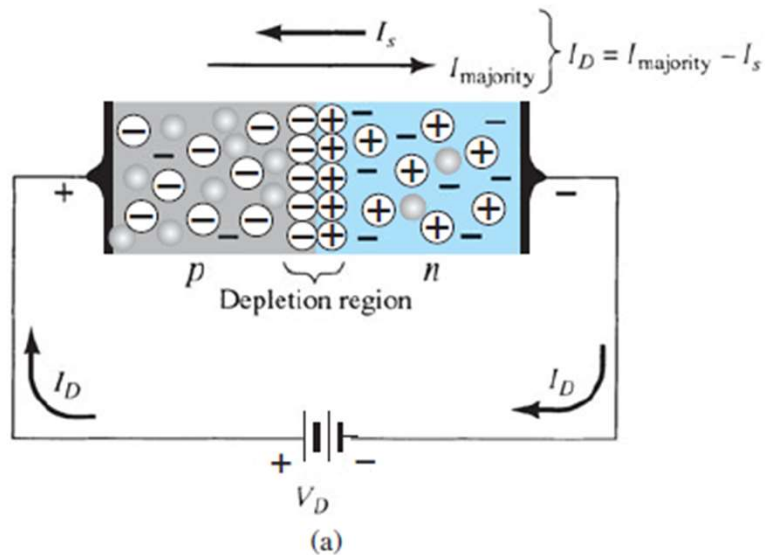
- Widening of the depletion region
 - No majority carriers flow.
 - Minority carriers flow same as before to generate the *reverse saturation current, I_s* .
 - Minority carriers flow increase with temperature .
- The reverse saturation current is seldom more than a few microamperes and typically in nA, except for high-power devices.
- it reaches its maximum level quickly and does not change significantly with increases in the reverse-bias potential



Semiconductor Diode Cont'd

Forward bias ($V_D > 0$)

- Depletion region width reduces.
- Minority carrier flow is same as before.
- Majority carrier flow increases until an exponential rise in current is achieved.



Shockley's equation

$$I_D = I_s(e^{V_D/nV_T} - 1) \quad (\text{A}) \quad (1.2)$$

where I_s is the reverse saturation current
 V_D is the applied forward-bias voltage across the diode
 n is an ideality factor, which is a function of the operating conditions and physical construction; it has a range between 1 and 2 depending on a wide variety of factors ($n = 1$ will be assumed throughout this text unless otherwise noted).

The voltage V_T in Eq. (1.1) is called the *thermal voltage* and is determined by

$$V_T = \frac{kT_K}{q} \quad (\text{V}) \quad (1.3)$$

where k is Boltzmann's constant = 1.38×10^{-23} J/K
 T_K is the absolute temperature in kelvins = $273 +$ the temperature in $^{\circ}\text{C}$
 q is the magnitude of electronic charge = 1.6×10^{-19} C

Ideal Semiconductor Diode

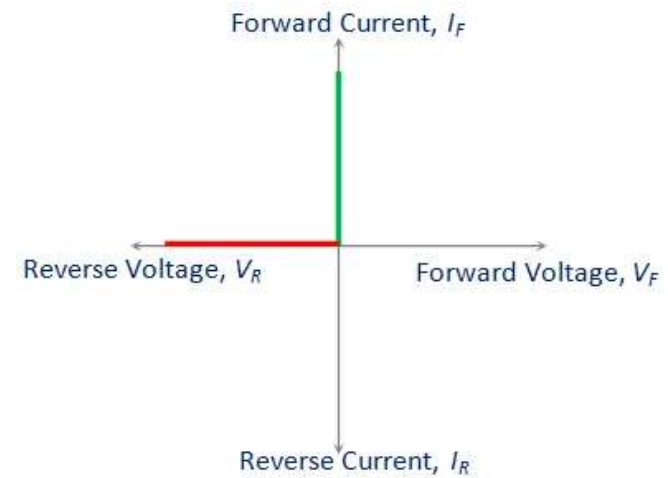
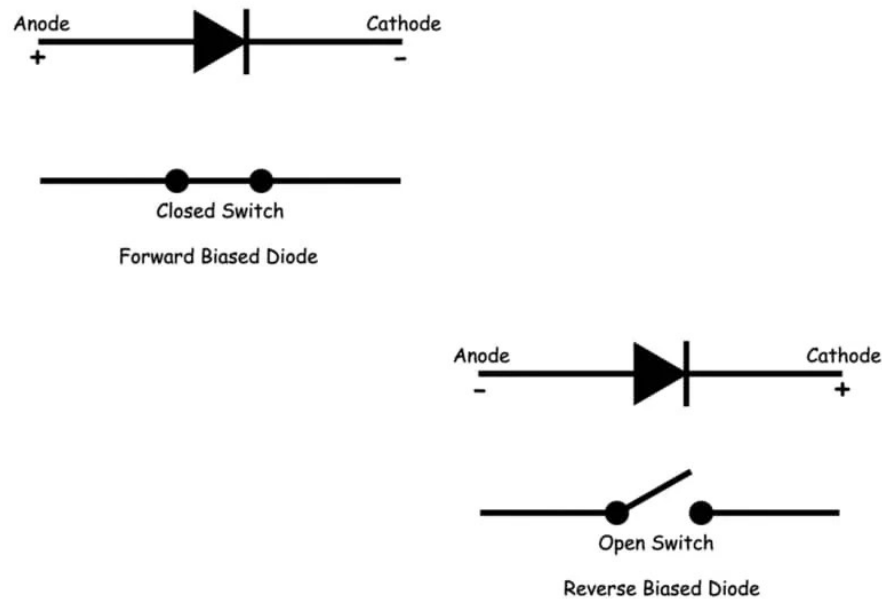
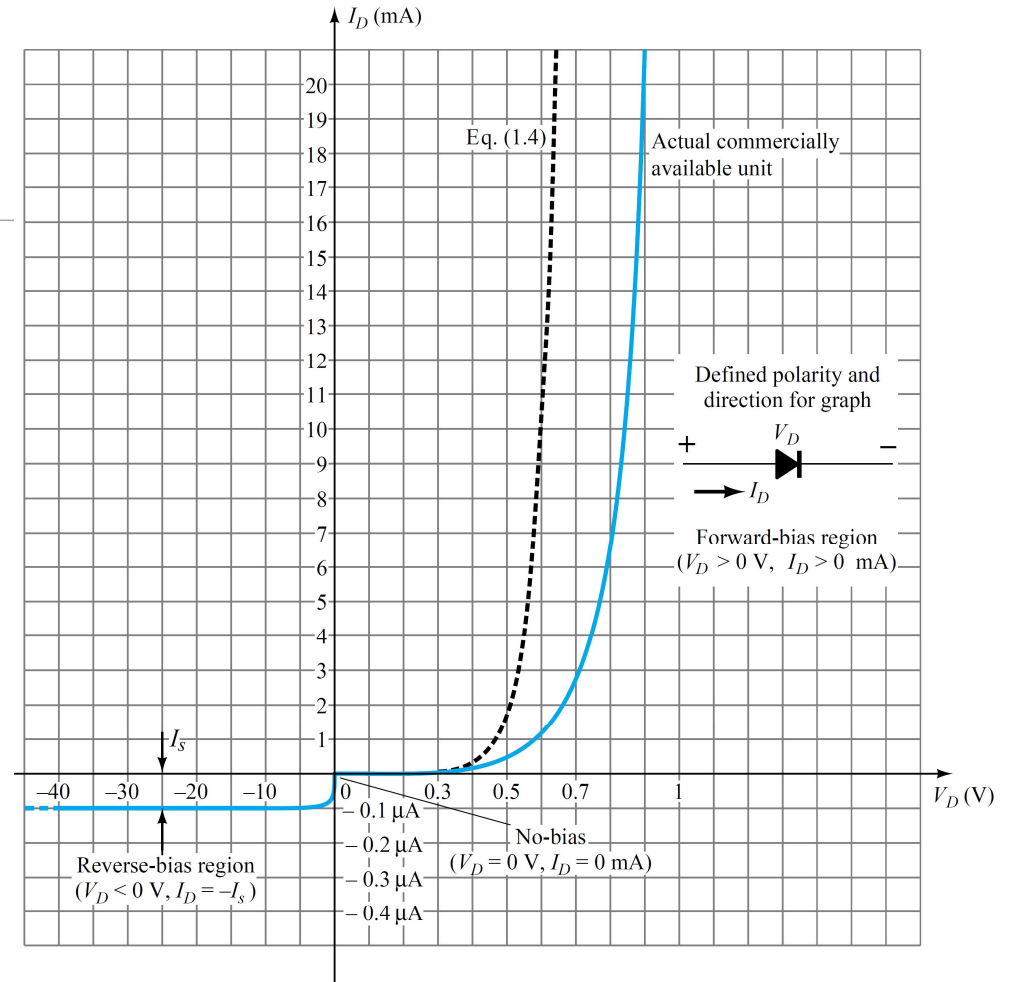


Figure 1 I-V Characteristics of an Ideal Diode

Actual Semiconductor Diode



Breakdown Region

The reverse-bias potential that results in this dramatic change in characteristics is called the breakdown potential and is given the label V_{BV} .

- As the voltage across the diode increases in the RV region, the velocity of the minority carriers responsible for the reverse saturation current I_s will increase.
- kinetic energy will be sufficient to release additional carriers through collisions with *otherwise stable atomic structures*.
- *ionization* process will result (whereby valence electrons absorb sufficient energy to leave the parent atom.)
- These additional carriers can then aid the ionization process to the point where a high *avalanche* current is established and the *avalanche breakdown* region determined.
- The avalanche region (V_{BV}) can be brought closer to the vertical **axis by increasing the doping levels**.

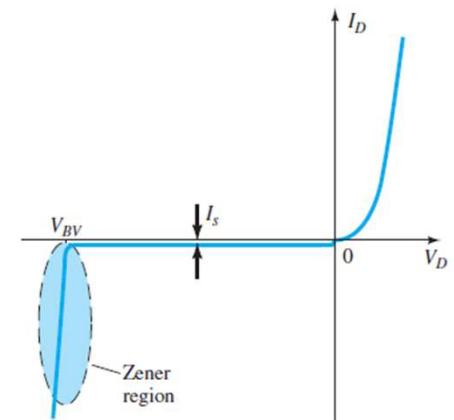


FIG. 1.17
Breakdown region.

-
- The maximum reverse-bias potential that can be applied before entering the breakdown region is called the **peak inverse voltage** (referred to simply as the PIV rating) or the peak reverse voltage (denoted the PRV rating).

EXAMPLE 1.1 At a temperature of 27°C (common temperature for components in an enclosed operating system), determine the thermal voltage V_T .

Solution: Substituting into Eq. (1.3), we obtain

$$\begin{aligned} T &= 273 + ^\circ\text{C} = 273 + 27 = 300 \text{ K} \\ V_T &= \frac{kT_K}{q} = \frac{(1.38 \times 10^{-23} \text{ J/K})(300 \text{ K})}{1.6 \times 10^{-19} \text{ C}} \\ &= 25.875 \text{ mV} \cong 26 \text{ mV} \end{aligned}$$

The thermal voltage will become an important parameter in the analysis to follow in this chapter and a number of those to follow.

RESISTANCE LEVELS (DC or Static Resistance AND AC or Dynamic Resistance)

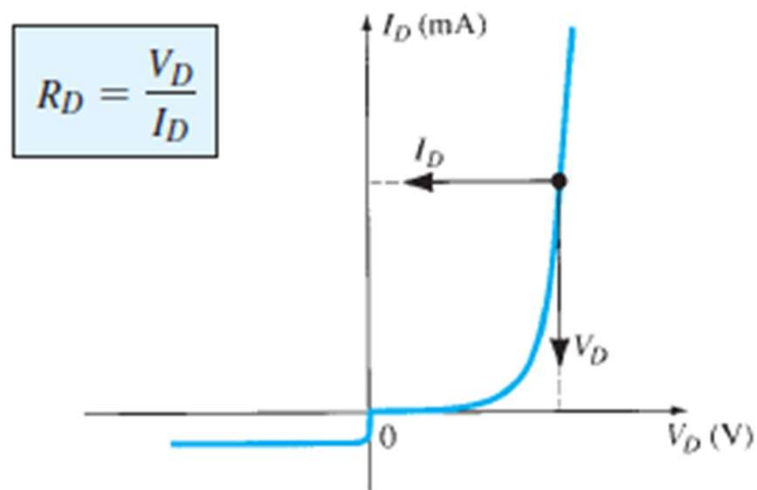


FIG. 1.23
Determining the dc resistance of a diode at a particular operating point.

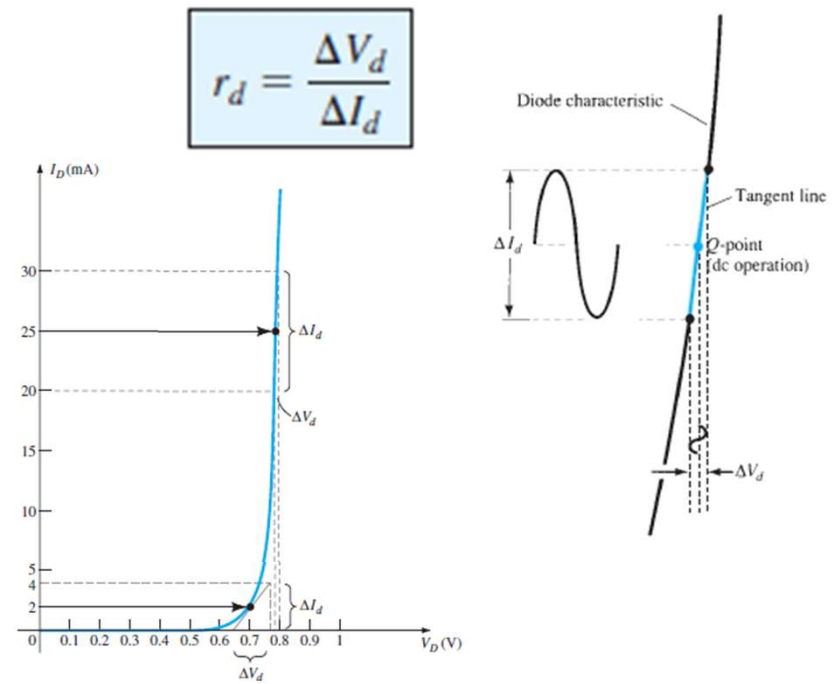


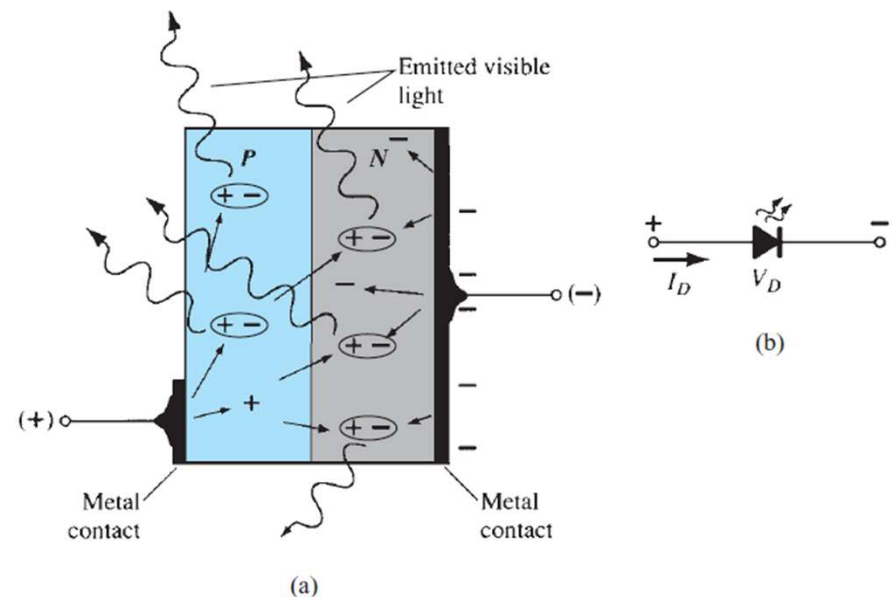
FIG. 1.27
Example 1.4.

LED(light-emitting diode)

- Not seen in Si, Ge
- GaAs emit infrared light .
- forward-biased, electrons cross the pn junction from the n -type material and recombine with holes in the p -type material
- these free electrons are in the conduction band and at a higher energy than the holes in the valence band.
- The difference in energy between the electrons and the holes corresponds to the energy of visible light.
- When recombination takes place, the recombining electrons release energy in the form of **photons**.

Light-Emitting Diodes

Color	Construction	Typical Forward Voltage (V)
Amber	AlInGaP	2.1
Blue	GaN	5.0
Green	GaP	2.2
Orange	GaAsP	2.0
Red	GaAsP	1.8
White	GaN	4.1
Yellow	AlInGaP	2.1



Check!

- Diode Bias
- Diode contact potential

EXAMPLE 2.4 For the series diode configuration of Fig. 2.13, determine V_D , V_R , and I_D .

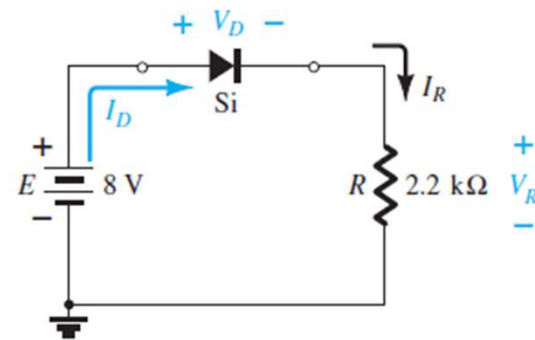


FIG. 2.13
Circuit for Example 2.4.

EXAMPLE 2.4 For the series diode configuration of Fig. 2.13, determine V_D , V_R , and I_D .

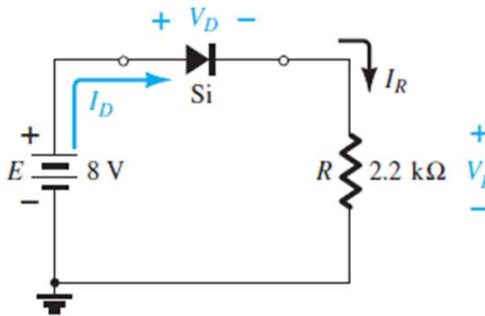


FIG. 2.13

Circuit for Example 2.4.

Solution: Since the applied voltage establishes a current in the clockwise direction to match the arrow of the symbol and the diode is in the “on” state,

$$V_D = 0.7\text{ V}$$

$$V_R = E - V_D = 8\text{ V} - 0.7\text{ V} = 7.3\text{ V}$$

$$I_D = I_R = \frac{V_R}{R} = \frac{7.3\text{ V}}{2.2\text{ k}\Omega} \cong 3.32\text{ mA}$$

Repeat with the diode reversed.

EXAMPLE 2.5 Repeat Example 2.4 with the diode reversed.

Solution: Removing the diode, we find that the direction of I is opposite to the arrow in the diode symbol and the diode equivalent is the open circuit no matter which model is employed. The result is the network of Fig. 2.14, where $I_D = 0$ A due to the open circuit. Since $V_R = I_R R$, we have $V_R = (0)R = 0$ V. Applying Kirchhoff's voltage law around the closed loop yields

$$E - V_D - V_R = 0$$

and

$$V_D = E - V_R = E - 0 = E = 8 \text{ V}$$

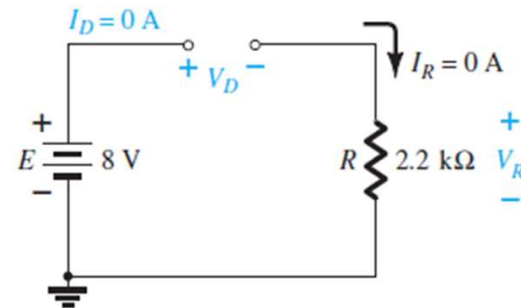


FIG. 2.14

Determining the unknown quantities for Example 2.5.

EXAMPLE 2.8 Determine I_D , V_{D_2} , and V_o for the circuit of Fig. 2.21.

Solution: Removing the diodes and determining the direction of the resulting current I result in the circuit of Fig. 2.22. There is a match in current direction for one silicon diode but not for the other silicon diode. The combination of a short circuit in series with an open circuit always results in an open circuit and $I_D = 0$ A, as shown in Fig. 2.23.

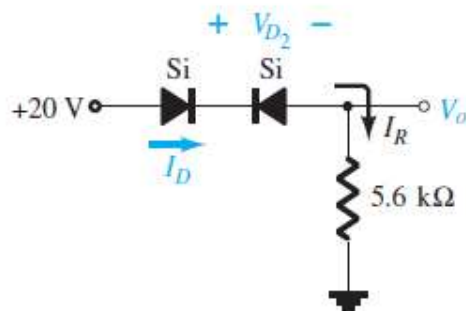


FIG. 2.21
Circuit for Example 2.8.

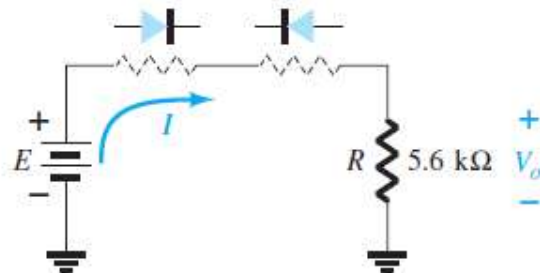


FIG. 2.22
Determining the state of the diodes
of Fig. 2.21.

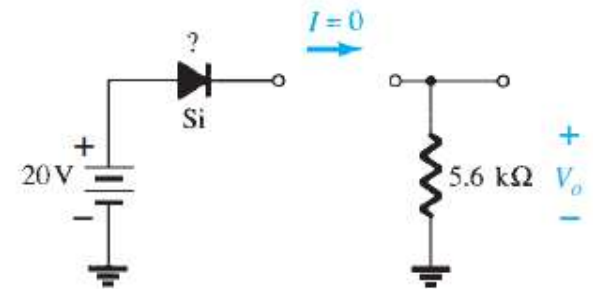


FIG. 2.23
Substituting the equivalent state for
the open diode.

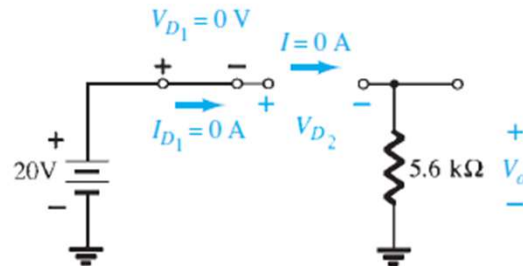


FIG. 2.24

Determining the unknown quantities for the circuit of Example 2.8.

The question remains as to what to substitute for the silicon diode. For the analysis to follow in this and succeeding chapters, simply recall for the actual practical diode that when $I_D = 0$ A, $V_D = 0$ V (and vice versa), as described for the no-bias situation in Chapter 1. The conditions described by $I_D = 0$ A and $V_{D1} = 0$ V are indicated in Fig. 2.24. We have

$$V_o = I_R R = I_D R = (0 \text{ A}) R = 0 \text{ V}$$

and

$$V_{D2} = V_{\text{open circuit}} = E = 20 \text{ V}$$

Applying Kirchhoff's voltage law in a clockwise direction gives

$$E - V_{D1} - V_{D2} - V_o = 0$$

and

$$\begin{aligned} V_{D2} &= E - V_{D1} - V_o = 20 \text{ V} - 0 - 0 \\ &= 20 \text{ V} \end{aligned}$$

with

$$V_o = 0 \text{ V}$$

EXAMPLE 2.9 Determine I , V_1 , V_2 , and V_o for the series dc configuration of Fig. 2.25.

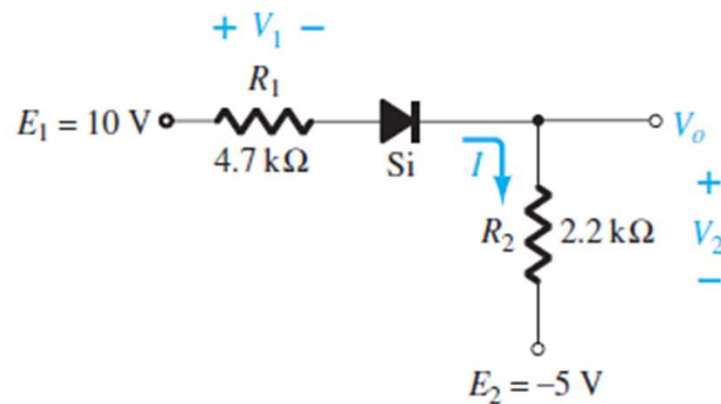


FIG. 2.25
Circuit for Example 2.9.

Solution: The sources are drawn and the current direction indicated as shown in Fig. 2.26. The diode is in the “on” state and the notation appearing in Fig. 2.27 is included to indicate this state. Note that the “on” state is noted simply by the additional $V_D = 0.7$ V on the figure. This eliminates the need to redraw the network and avoids any confusion that may

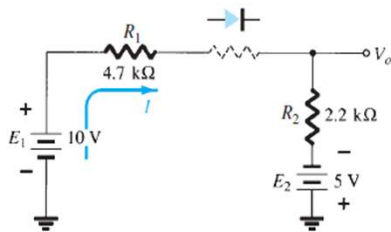


FIG. 2.26

Determining the state of the diode for the network of Fig. 2.25.

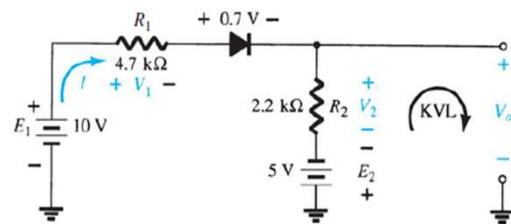


FIG. 2.27

Determining the unknown quantities for the network of Fig. 2.25. KVL, Kirchhoff voltage loop.

result from the appearance of another source. As indicated in the introduction to this section, this is probably the path and notation that one will take when a level of confidence has been established in the analysis of diode configurations. In time the entire analysis will be performed simply by referring to the original network. Recall that a reverse-biased diode can simply be indicated by a line through the device.

The resulting current through the circuit is

$$I = \frac{E_1 + E_2 - V_D}{R_1 + R_2} = \frac{10 \text{ V} + 5 \text{ V} - 0.7 \text{ V}}{4.7 \text{ k}\Omega + 2.2 \text{ k}\Omega} = \frac{14.3 \text{ V}}{6.9 \text{ k}\Omega} \cong 2.07 \text{ mA}$$

and the voltages are

$$V_1 = IR_1 = (2.07 \text{ mA})(4.7 \text{ k}\Omega) = 9.73 \text{ V}$$

$$V_2 = IR_2 = (2.07 \text{ mA})(2.2 \text{ k}\Omega) = 4.55 \text{ V}$$

Applying Kirchhoff's voltage law to the output section in the clockwise direction results in

$$-E_2 + V_2 - V_o = 0$$

and

$$V_o = V_2 - E_2 = 4.55 \text{ V} - 5 \text{ V} = -0.45 \text{ V}$$

The minus sign indicates that V_o has a polarity opposite to that appearing in Fig. 2.25.

Practice Yourself Example – 2.10(Boyelstad)

SINUSOIDAL INPUTS; HALF-WAVE RECTIFICATION

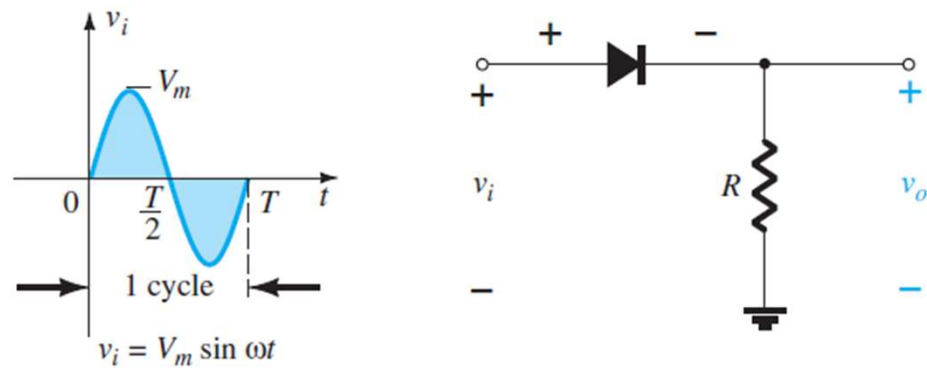


FIG. 2.44
Half-wave rectifier.

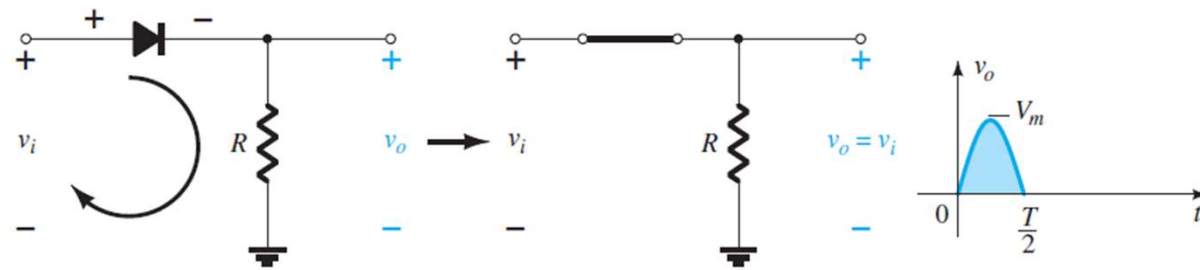


FIG. 2.45

Conduction region ($0 \rightarrow T/2$).

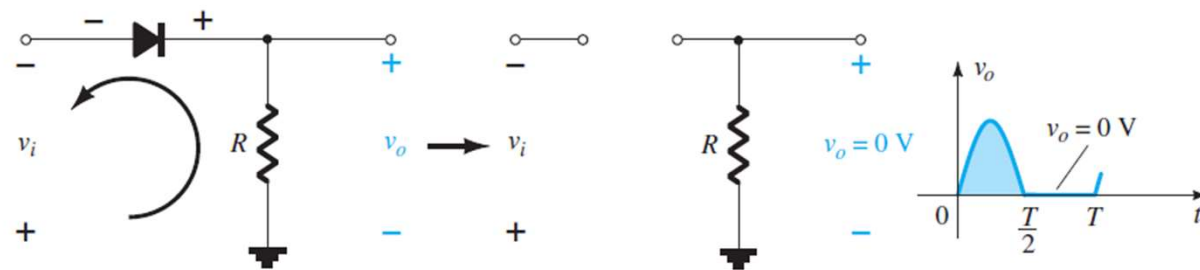


FIG. 2.46

Nonconduction region ($T/2 \rightarrow T$).

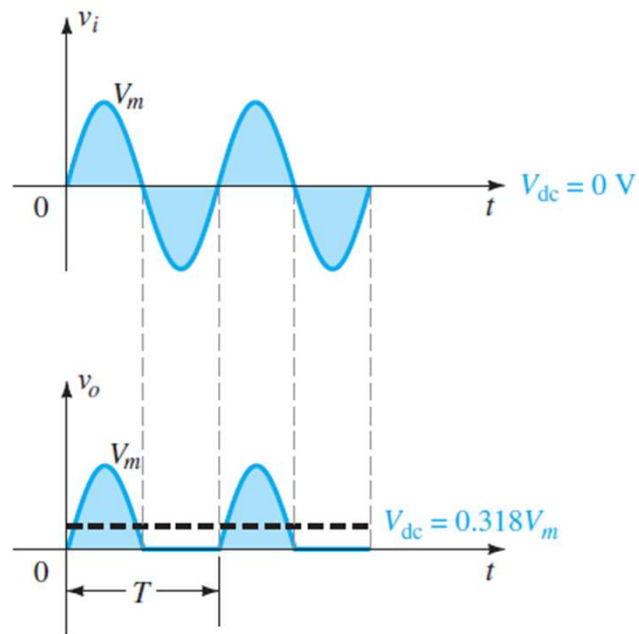


FIG. 2.47
Half-wave rectified signal.

a full period and an average value determined by

$$V_{dc} = 0.318 V_m \quad \text{half-wave}$$

The process of removing one-half the input signal to establish a dc level is called *half-wave rectification*.

The effect of using a silicon diode with $V_K = 0.7 \text{ V}$ is demonstrated in Fig. 2.48 for the forward-bias region. The applied signal must now be at least 0.7 V before the diode can turn “on.” For levels of v_i less than 0.7 V , the diode is still in an open-circuit state and $v_o = 0 \text{ V}$, as shown in the same figure. When conducting, the difference between v_o and v_i is a fixed

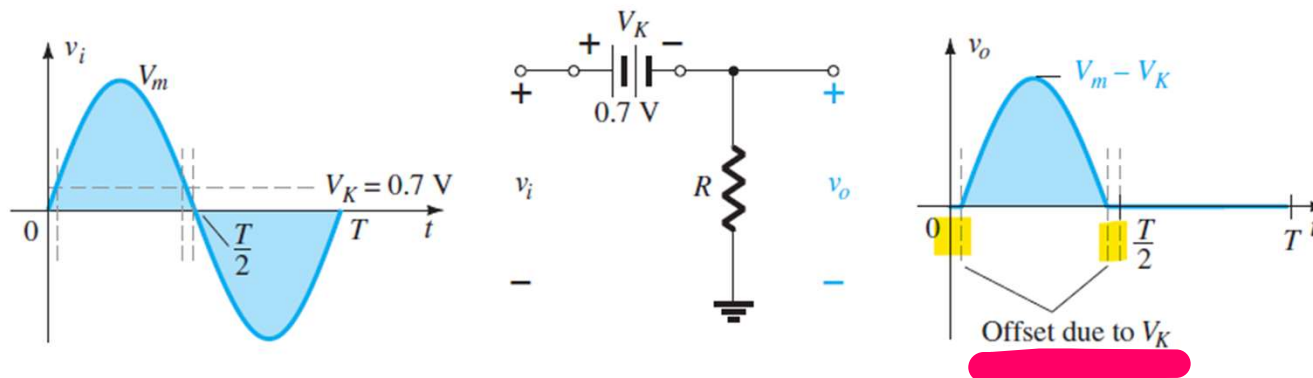


FIG. 2.48

Effect of V_K on half-wave rectified signal.

$$V_{dc} \cong 0.318(V_m - V_K)$$

PIV (PRV)

The peak inverse voltage (PIV) [or PRV (peak reverse voltage)] rating of the diode is of primary importance in the design of rectification systems. Recall that it is the voltage rating that must not be exceeded in the reverse-bias region or the diode will enter the Zener avalanche region. The required PIV rating for the half-wave rectifier can be determined from Fig. 2.52, which displays the reverse-biased diode of Fig. 2.44 with maximum applied voltage. Applying Kirchhoff's voltage law, it is fairly obvious that the PIV rating of the diode must equal or exceed the peak value of the applied voltage. Therefore,

$$\text{PIV rating} \geq V_m \quad \text{half-wave rectifier} \quad (2.9)$$

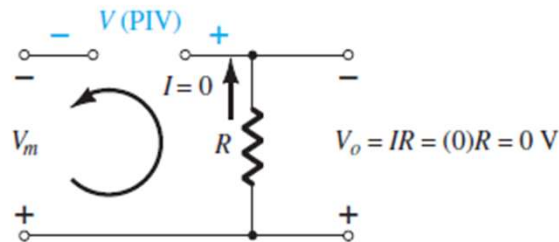


FIG. 2.52

Determining the required PIV rating for the half-wave rectifier.

FULL-WAVE RECTIFICATION

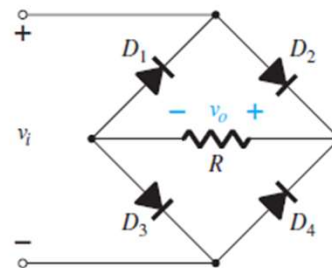
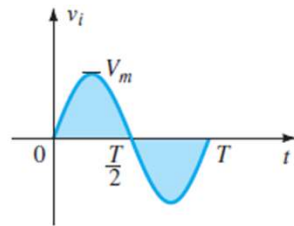


FIG. 2.53

Full-wave bridge rectifier.

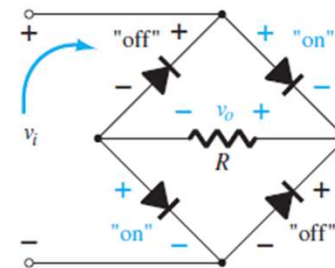


FIG. 2.54

Network of Fig. 2.53 for the period $0 \rightarrow T/2$ of the input voltage v_i .

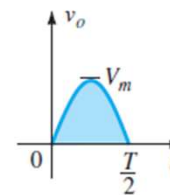
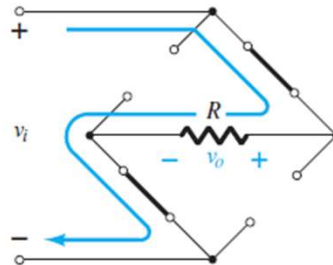
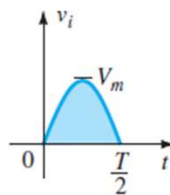


FIG. 2.55

Conduction path for the positive region of v_i .

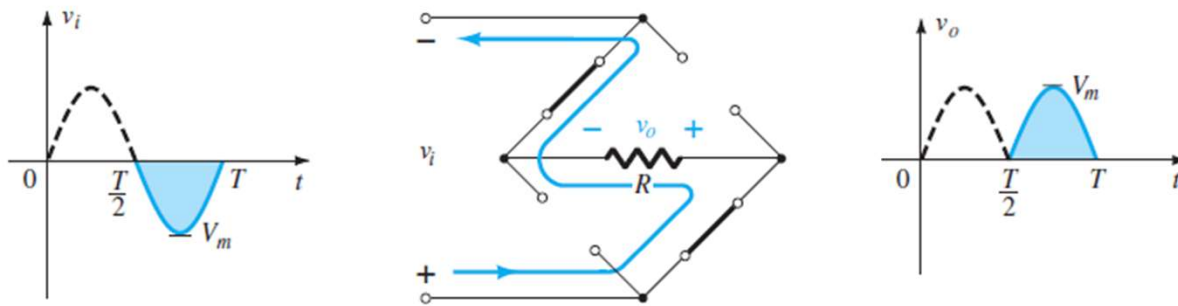


FIG. 2.56

Conduction path for the negative region of v_i .

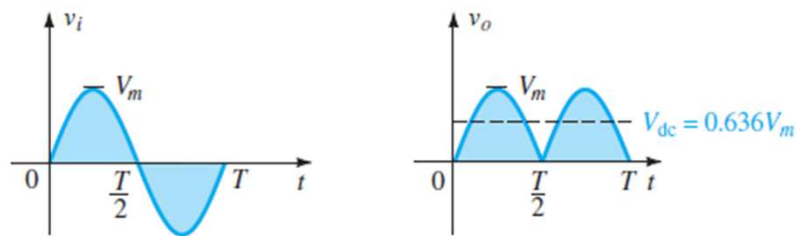


FIG. 2.57

Input and output waveforms for a full-wave rectifier.

$$V_{dc} = 0.636 V_m$$

full-wave

For Silicon diodes

If silicon rather than ideal diodes are employed as shown in Fig. 2.58, the application of Kirchhoff's voltage law around the conduction path results in

$$v_i - V_K - v_o - V_K = 0$$

and

$$v_o = v_i - 2V_K$$

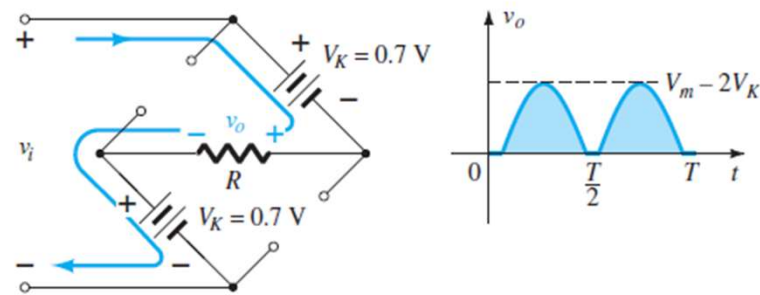


FIG. 2.58

Determining $V_{o_{\max}}$ for silicon diodes in the bridge configuration.

The peak value of the output voltage v_o is therefore

$$V_{o_{\max}} = V_m - 2V_K$$

For situations where $V_m \gg 2V_K$, the following equation can be applied for the average value with a relatively high level of accuracy:

$$V_{dc} \cong 0.636(V_m - 2V_K) \quad (2.11)$$

PIV The required PIV of each diode (ideal) can be determined from Fig. 2.59 obtained at the peak of the positive region of the input signal. For the indicated loop the maximum voltage across R is V_m and the PIV rating is defined by

$$\text{PIV} \cong V_m \quad \text{full-wave bridge rectifier} \quad (2.12)$$

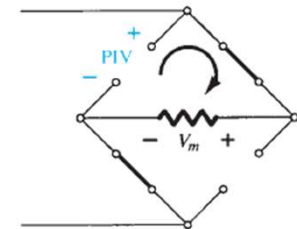


FIG. 2.59
Determining the required PIV for the bridge configuration.

Center-Tapped Transformer

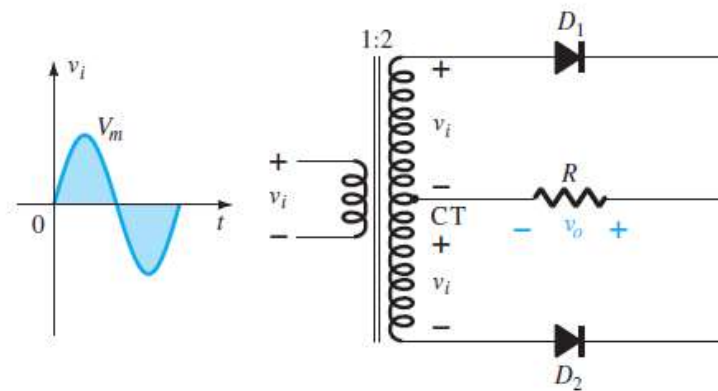


FIG. 2.60

Center-tapped transformer full-wave rectifier.

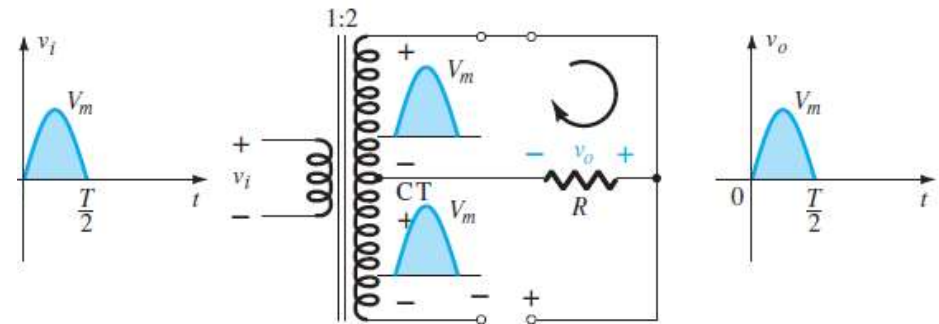


FIG. 2.61

Network conditions for the positive region of v_i .

During the negative portion of the input the network appears as shown in Fig. 2.62, reversing the roles of the diodes but maintaining the same polarity for the voltage across the load resistor R . The net effect is the same output as that appearing in Fig. 2.57 with the same dc levels.

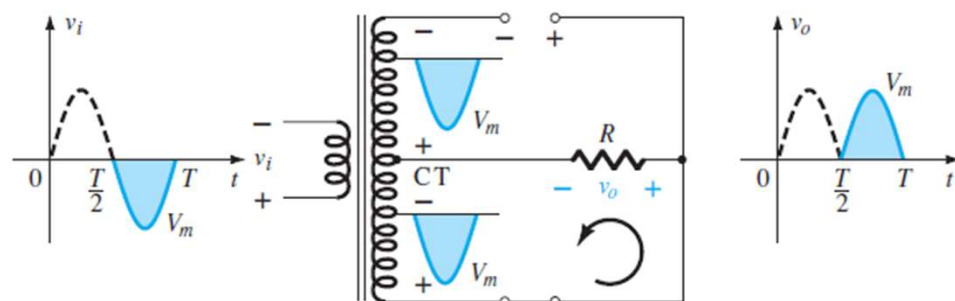


FIG. 2.62

Network conditions for the negative region of v_i .

PIV The network of Fig. 2.63 will help us determine the net PIV for each diode for this full-wave rectifier. Inserting the maximum voltage for the secondary voltage and V_m as established by the adjoining loop results in

$$\begin{aligned} \text{PIV} &= V_{\text{secondary}} + V_R \\ &= V_m + V_m \end{aligned}$$

and

$$\boxed{\text{PIV} \cong 2V_m}$$

CT transformer, full-wave rectifier

(2.13)

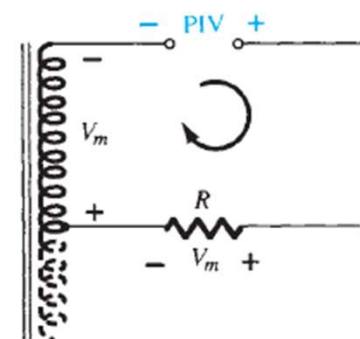


FIG. 2.63

Determining the PIV level for the diodes of the CT transformer full-wave rectifier.

EXAMPLE 2.17 Determine the output waveform for the network of Fig. 2.64 and calculate the output dc level and the required PIV of each diode.

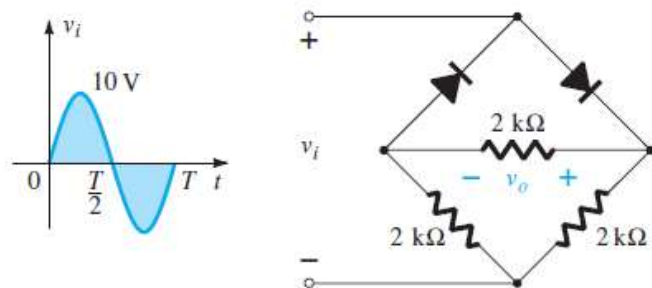


FIG. 2.64
Bridge network for Example 2.17.

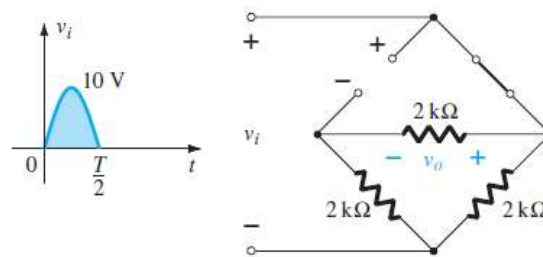


FIG. 2.65
Network of Fig. 2.64 for the positive region of v_i .

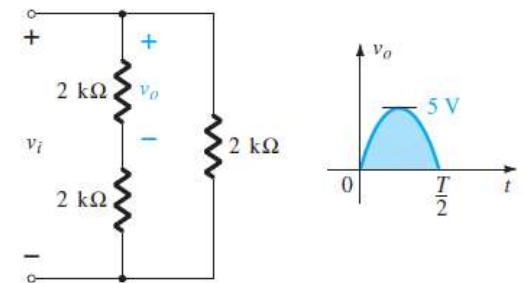


FIG. 2.66
Redrawn network of Fig. 2.65.

Solution: The network appears as shown in Fig. 2.65 for the positive region of the input voltage. Redrawing the network results in the configuration of Fig. 2.66, where $v_o = \frac{1}{2}v_i$ or $V_{o_{\max}} = \frac{1}{2}V_{i_{\max}} = \frac{1}{2}(10 \text{ V}) = 5 \text{ V}$, as shown in Fig. 2.66. For the negative part of the input, the roles of the diodes are interchanged and v_o appears as shown in Fig. 2.67.

The effect of removing two diodes from the bridge configuration is therefore to reduce the available dc level to the following:

$$V_{\text{dc}} = 0.636(5 \text{ V}) = 3.18 \text{ V}$$

or that available from a half-wave rectifier with the same input. However, the PIV as determined from Fig. 2.59 is equal to the maximum voltage across R , which is 5 V, or half of that required for a half-wave rectifier with the same input.

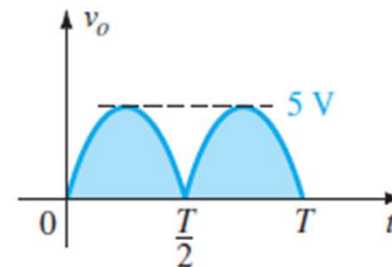


FIG. 2.67

Resulting output for Example 2.17.

CLIPPERS

- Clippers are networks that employ diodes to “clip” away a portion of an input signal without distorting the remaining part of the applied waveform.
- There are two general categories of clippers: series and parallel.
- The series configuration is defined as one where the diode is in series with the load, whereas
- the parallel variety has the diode in a branch parallel to the load.

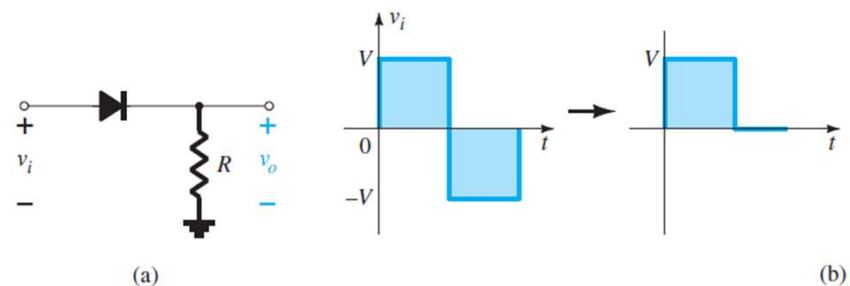
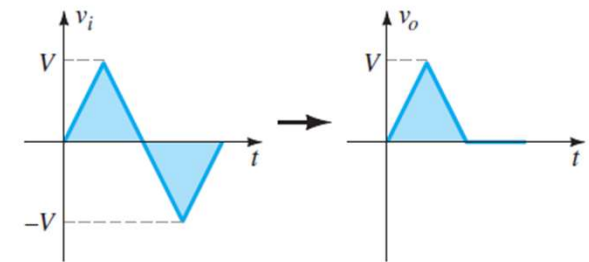


FIG. 2.68
Series clipper.

SERIES CLIPPERS

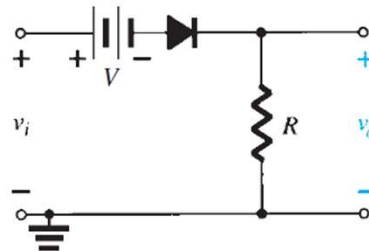
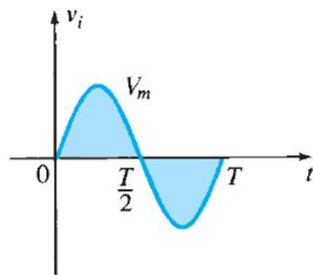


FIG. 2.69

Series clipper with a dc supply.

- the dc supply can aid or work against the source voltage!
- determine the applied voltage (transition voltage)

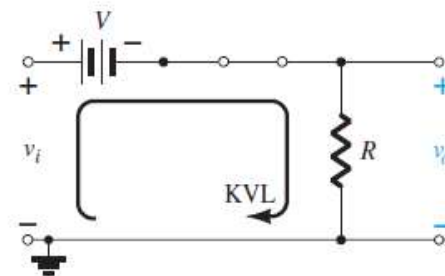


FIG. 2.72

Determining v_o for the diode in the "on" state.

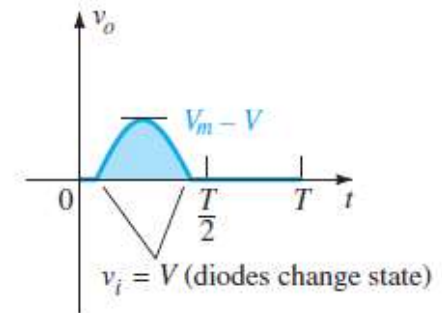
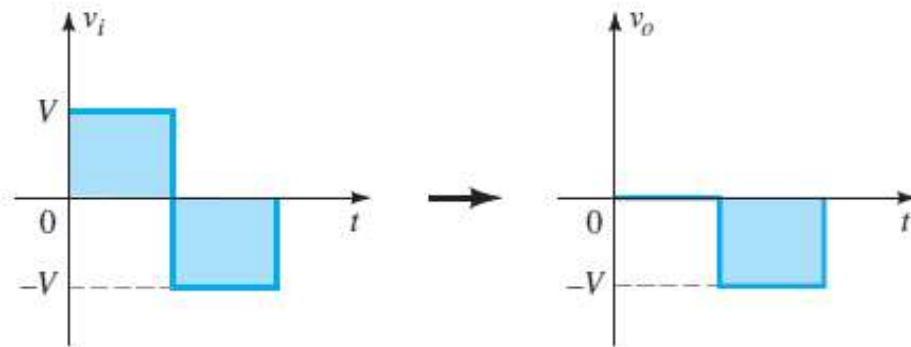
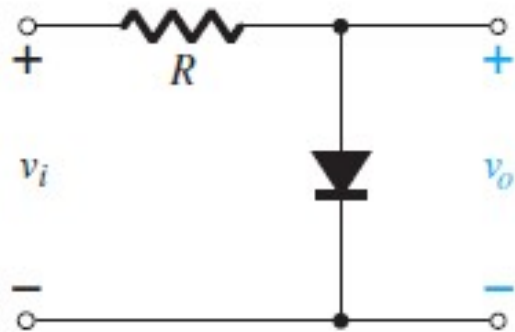


FIG. 2.73

Sketching the waveform of v_o using the results obtained for v_o above and below the transition level.

PARALLEL CLIPPERS



EXAMPLE 2.20 Determine v_o for the network of Fig. 2.82.

Solution:

Step 1: In this example the output is defined across the series combination of the 4-V supply and the diode, not across the resistor R .

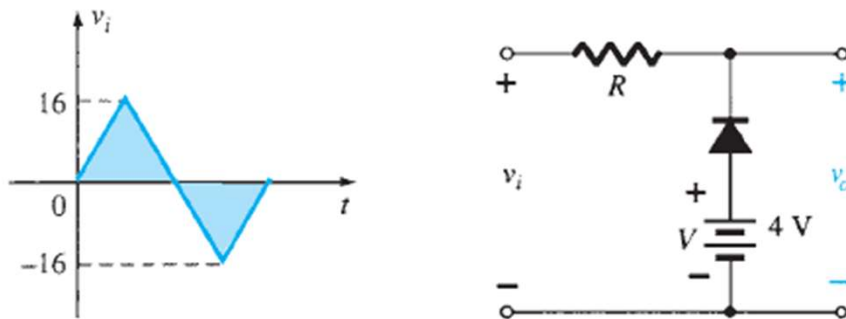


FIG. 2.82
Example 2.20.

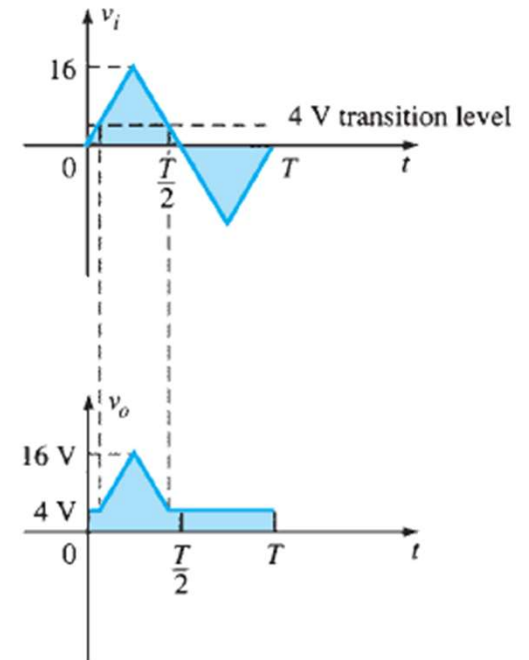


FIG. 2.84
Sketching v_o for Example 2.20.

Positive half cycle –

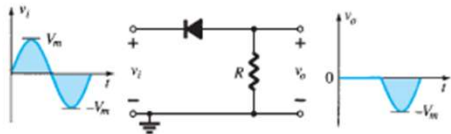
Diode is

- FB at voltage < 4 V
- reversed biased for voltage > 4 V

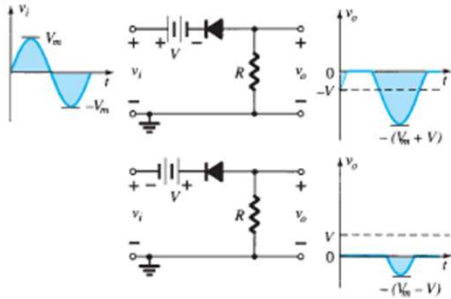
Practice similar problems

Simple Series Clippers (Ideal Diodes)

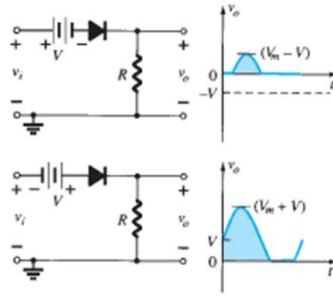
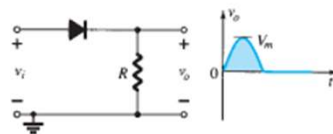
POSITIVE



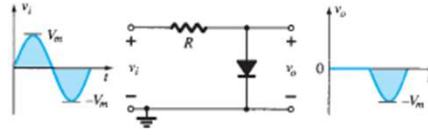
Biased Series Clippers (Ideal Diodes)



NEGATIVE



Simple Parallel Clippers (Ideal Diodes)



Biased Parallel Clippers (Ideal Diodes)

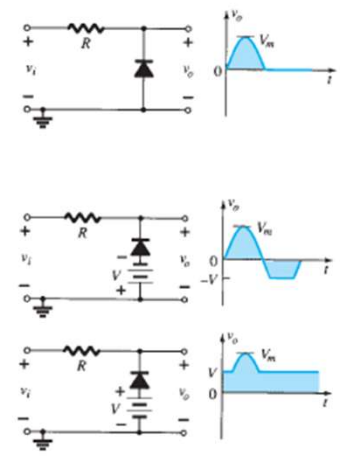
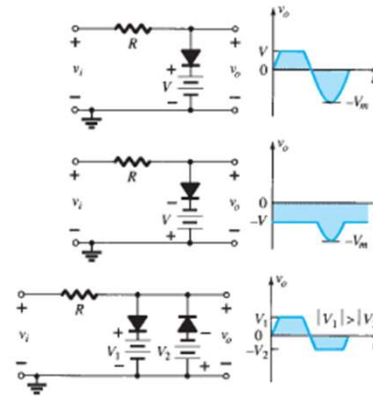


FIG. 2.88
Clipping circuits.

CLAMPERS

- A clamper is a network constructed of a diode, a resistor, and a capacitor that shifts a waveform to a different dc level without changing the appearance of the applied signal.

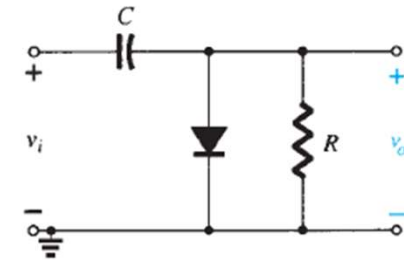
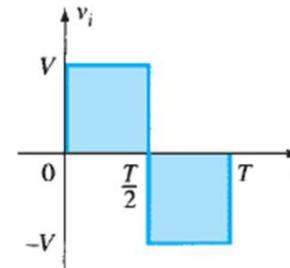


FIG. 2.89
Clamper.

Step 1: Start the analysis by examining the response of the portion of the input signal that will forward bias the diode.

Step 2: During the period that the diode is in the “on” state, assume that the capacitor will charge up instantaneously to a voltage level determined by the surrounding network.

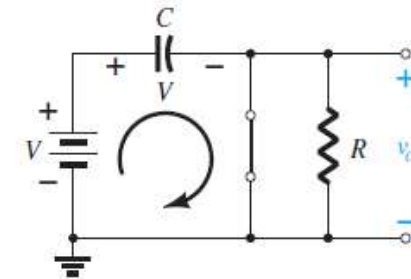


FIG. 2.90
Diode “on” and the capacitor charging to V volts.

Step 3: Assume that during the period when the diode is in the “off” state the capacitor holds on to its established voltage level.

Step 4: Throughout the analysis, maintain a continual awareness of the location and defined polarity for v_o to ensure that the proper levels are obtained.

- The chosen resistor and capacitor of the network must be chosen such that the time constant determined by $\tau = RC$ is sufficiently large to ensure that the voltage across the capacitor does not discharge significantly during the interval the diode is nonconducting. Throughout the analysis we assume that for all practical purposes the capacitor fully charges or discharges in five time constants.

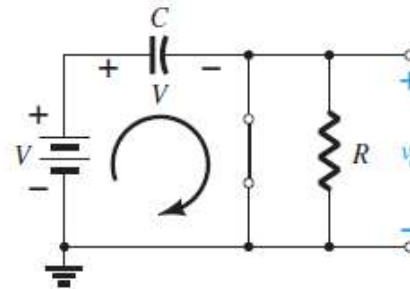


FIG. 2.90
Diode “on” and the capacitor charging to V volts.

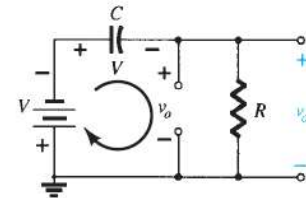


FIG. 2.91
Determining v_o with the diode “off.”

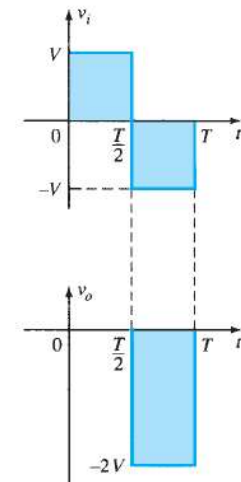


FIG. 2.92
Sketching v_o for the network of Fig. 2.91.

EXAMPLE 2.22 Determine v_o for the network of Fig. 2.93 for the input indicated.

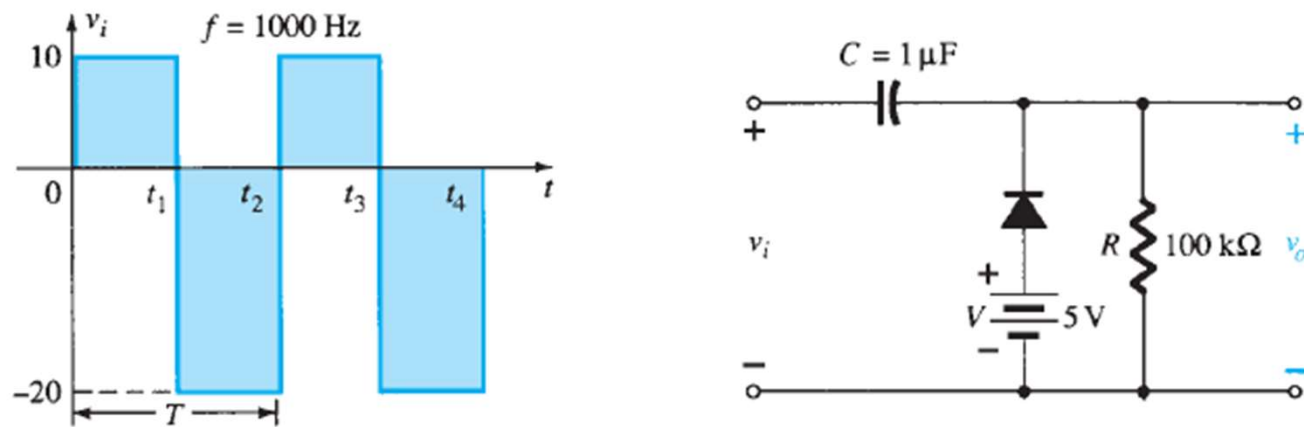


FIG. 2.93

Applied signal and network for Example 2.22.

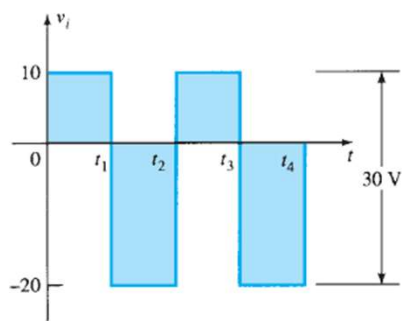
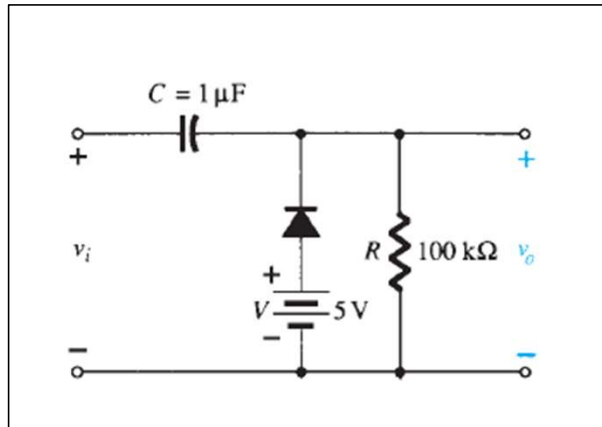


FIG. 2.96

v_i and v_o for the clamper of Fig. 2.93.

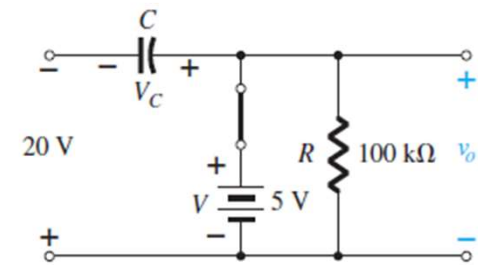
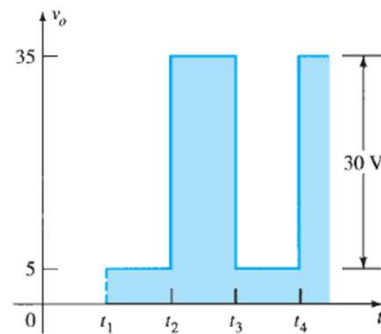


FIG. 2.94

Determining v_o and V_C with the diode in the "on" state.

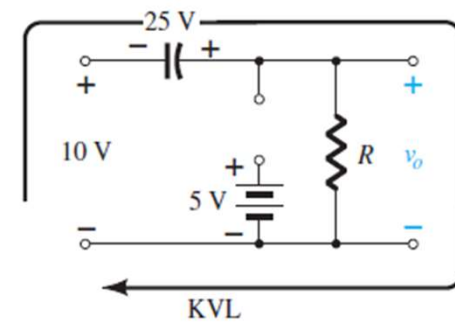


FIG. 2.95

Determining v_o with the diode in the "off" state.

Practice Yourself Example 2.23

Zener Voltage, Zener Breakdown and Zener Region

Instead of a straight line representing the cathode, the zener diode has a bent line that reminds you of the letter Z (for zener).

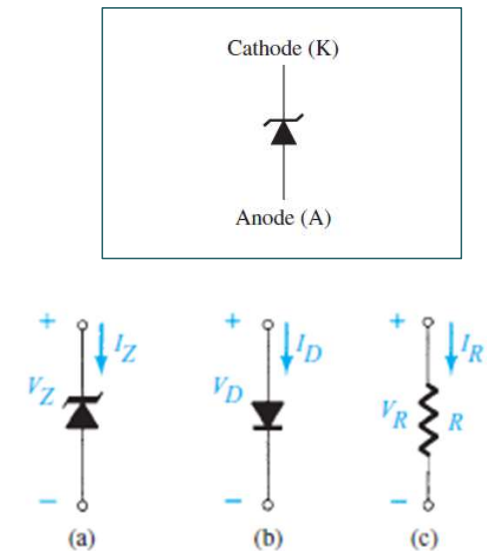
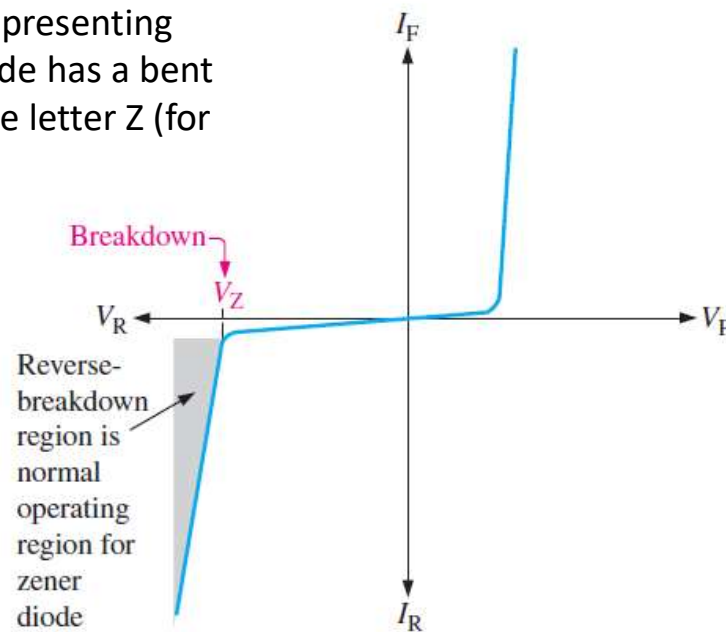
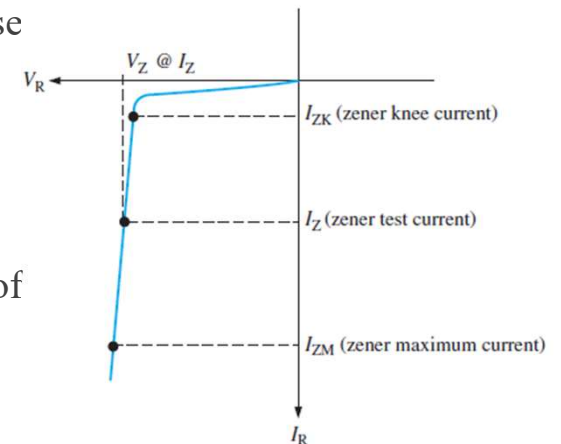


FIG. 1.46

Conduction direction: (a) Zener diode;
(b) semiconductor diode;
(c) resistive element.

Zener Voltage, Zener Breakdown and Zener Region

- use breakdown region characteristics of the p-n junction diode.(operate in reverse breakdown)
- A zener diode is heavily doped to reduce the breakdown voltage.
- Thinner depletion region .
- The slight slope to the curve in the Zener region reveals that there is a level of resistance to be associated with the Zener diode in the conduction mode.
- The location of the Zener region can be controlled by varying the doping levels.
- The ability to keep the reverse voltage across its terminals essentially constant is the key feature of the zener diode.
- A zener diode operating in breakdown acts as a voltage regulator.



Zener Voltage, Zener Breakdown and Zener Region

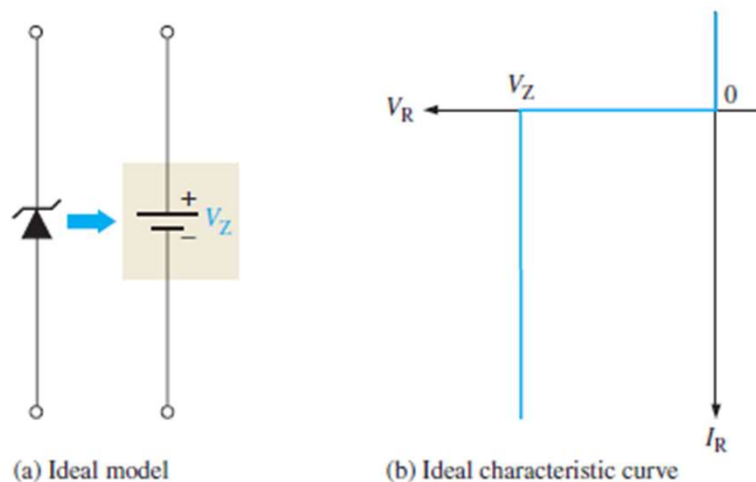
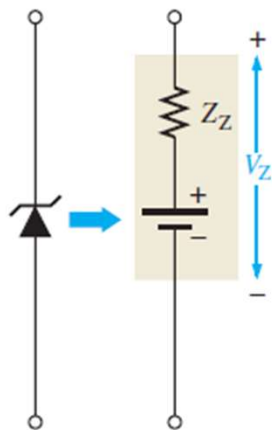
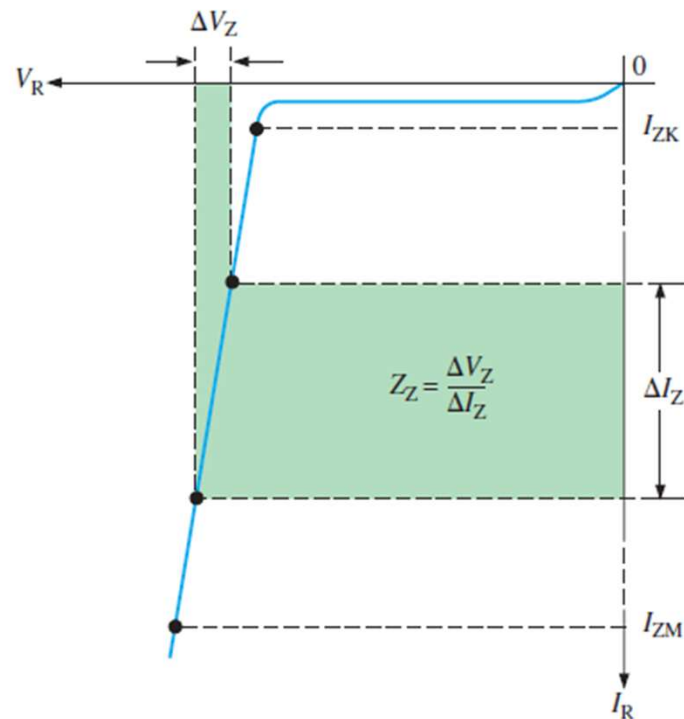


FIGURE 3-4
Ideal zener diode equivalent circuit model and the characteristic curve.



(a) Practical model



(b) Characteristic curve. The slope is exaggerated for illustration.

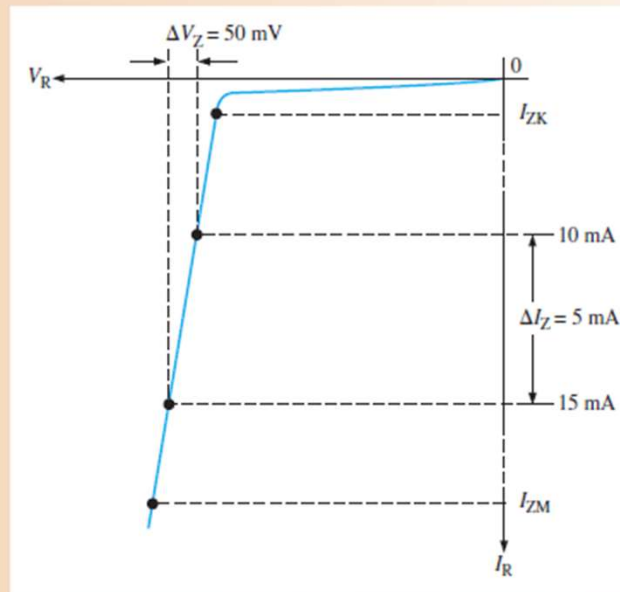
◀ FIGURE 3-5

Practical zener diode equivalent circuit and the characteristic curve illustrating Z_Z .

EXAMPLE 3-1

A zener diode exhibits a certain change in V_Z for a certain change in I_Z on a portion of the linear characteristic curve between I_{ZK} and I_{ZM} as illustrated in Figure 3-6. What is the zener impedance?

► FIGURE 3-6



Solution

$$Z_Z = \frac{\Delta V_Z}{\Delta I_Z} = \frac{50 \text{ mV}}{5 \text{ mA}} = 10 \Omega$$

Related Problem*

Calculate the zener impedance if the change in zener voltage is 100 mV for a 20 mA change in zener current on the linear portion of the characteristic curve.

*Answers can be found at www.pearsonhighered.com/floyd.

Data sheet (a portion)



DO-41 Glass case
COLOR BAND DENOTES CATHODE

Absolute Maximum Ratings * $T_a = 25^\circ\text{C}$ unless otherwise noted

Symbol	Parameter	Value	Units
P_D	Power Dissipation @ $T_L \leq 50^\circ\text{C}$, Lead Length = 3/8"	1.0	W
	Derate above 50°C	6.67	mW/ $^\circ\text{C}$
T_J, T_{STG}	Operating and Storage Temperature Range	-65 to +200	$^\circ\text{C}$

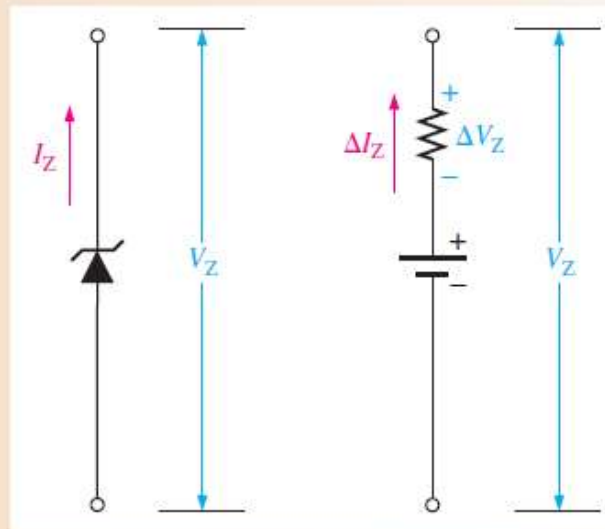
* These ratings are limiting values above which the serviceability of the diode may be impaired.

Electrical Characteristics $T_a = 25^\circ\text{C}$ unless otherwise noted

Device	V_Z (V) @ I_Z (Note 1)			Test Current I_Z (mA)	Max. Zener Impedance			Leakage Current	
	Min.	Typ.	Max.		Z_Z @ I_Z (Ω)	Z_{ZK} @ I_{ZK} (Ω)	I_{ZK} (mA)	I_R (μA)	V_R (V)
1N4728A	3.315	3.3	3.465	76	10	400	1	100	1
1N4729A	3.42	3.6	3.78	69	10	400	1	100	1
1N4730A	3.705	3.9	4.095	64	9	400	1	50	1
1N4731A	4.085	4.3	4.515	58	9	400	1	10	1
1N4732A	4.465	4.7	4.935	53	8	500	1	10	1
1N4733A	4.845	5.1	5.355	49	7	550	1	10	1
1N4734A	5.32	5.6	5.88	45	5	600	1	10	2
1N4735A	5.89	6.2	6.51	41	2	700	1	10	3
1N4736A	6.46	6.8	7.14	37	3.5	700	1	10	4
1N4737A	7.125	7.5	7.875	34	4	700	0.5	10	5

EXAMPLE 3-4

From the datasheet in Figure 3-7, a 1N4736A zener diode has a Z_Z of $3.5\ \Omega$. The datasheet gives $V_Z = 6.8\ \text{V}$ at a test current, I_Z , of 37 mA. What is the voltage across the zener terminals when the current is 50 mA? When the current is 25 mA? Figure 3-8 represents the zener diode.



▲ FIGURE 3-8

Solution For $I_Z = 50$ mA: The 50 mA current is a 13 mA increase above the test current, I_Z , of 37 mA.

$$\Delta I_Z = I_Z - 37 \text{ mA} = 50 \text{ mA} - 37 \text{ mA} = +13 \text{ mA}$$

$$\Delta V_Z = \Delta I_Z Z_Z = (13 \text{ mA})(3.5 \Omega) = +45.5 \text{ mV}$$

The change in voltage due to the increase in current above the I_Z value causes the zener terminal voltage to increase. The zener voltage for $I_Z = 50$ mA is

$$V_Z = 6.8 \text{ V} + \Delta V_Z = 6.8 \text{ V} + 45.5 \text{ mV} = \mathbf{6.85 \text{ V}}$$

For $I_Z = 25$ mA: The 25 mA current is a 12 mA decrease below the test current, I_Z , of 37 mA.

$$\Delta I_Z = -12 \text{ mA}$$

$$\Delta V_Z = \Delta I_Z Z_Z = (-12 \text{ mA})(3.5 \Omega) = -42 \text{ mV}$$

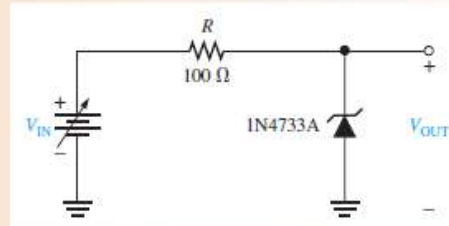
The change in voltage due to the decrease in current below the test current causes the zener terminal voltage to decrease. The zener voltage for $I_Z = 25$ mA is

$$V_Z = 6.8 \text{ V} - \Delta V_Z = 6.8 \text{ V} - 42 \text{ mV} = \mathbf{6.76 \text{ V}}$$

EXAMPLE 3-5

Determine the minimum and the maximum input voltages that can be regulated by the zener diode in Figure 3-11.

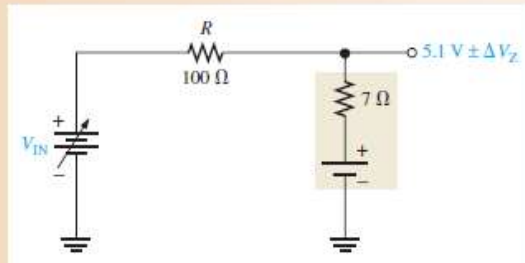
► **FIGURE 3-11**



Solution From the datasheet in Figure 3-7 for the 1N4733A: $V_Z = 5.1 \text{ V}$ at $I_Z = 49 \text{ mA}$, $I_{ZK} = 1 \text{ mA}$, and $Z_Z = 7 \Omega$ at I_Z . For simplicity, assume this value of Z_Z over the range of current values. The equivalent circuit is shown in Figure 3-12.

► **FIGURE 3-12**

Equivalent of circuit in Figure 3-11.



At $I_{ZK} = 1 \text{ mA}$, the output voltage is

$$\begin{aligned} V_{OUT} &\cong 5.1 \text{ V} - \Delta V_Z = 5.1 \text{ V} - (I_Z - I_{ZK})Z_Z = 5.1 \text{ V} - (49 \text{ mA} - 1 \text{ mA})(7 \Omega) \\ &= 5.1 \text{ V} - (48 \text{ mA})(7 \Omega) = 5.1 \text{ V} - 0.336 \text{ V} = 4.76 \text{ V} \end{aligned}$$

Therefore,

$$V_{IN(\min)} = I_{ZK}R + V_{OUT} = (1 \text{ mA})(100 \Omega) + 4.76 \text{ V} = 4.86 \text{ V}$$

To find the maximum input voltage, first calculate the maximum zener current. Assume the temperature is 50°C or below; so from Figure 3-7, the power dissipation is 1 W .

$$I_{ZM} = \frac{P_{D(\max)}}{V_Z} = \frac{1 \text{ W}}{5.1 \text{ V}} = 196 \text{ mA}$$

At I_{ZM} , the output voltage is

$$\begin{aligned} V_{OUT} &\cong 5.1 \text{ V} + \Delta V_Z = 5.1 \text{ V} + (I_{ZM} - I_Z)Z_Z \\ &= 5.1 \text{ V} + (196 \text{ mA} - 49 \text{ mA})(7 \Omega) = 5.1 \text{ V} + 1.03 \text{ V} = 6.13 \text{ V} \end{aligned}$$

Therefore,

$$V_{IN(\max)} = I_{ZM}R + V_{OUT} = (196 \text{ mA})(100 \Omega) + 6.13 \text{ V} = 25.7 \text{ V}$$

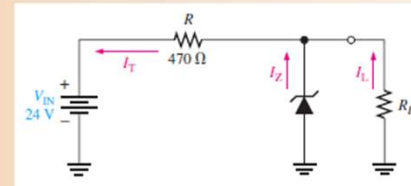
Zener Regulation with a Variable Load

EXAMPLE 3–6

Determine the minimum and the maximum load currents for which the zener diode in Figure 3–14 will maintain regulation. What is the minimum value of R_L that can be used? $V_Z = 12\text{ V}$, $I_{ZK} = 1\text{ mA}$, and $I_{ZM} = 50\text{ mA}$. Assume an ideal zener diode where $Z_Z = 0\ \Omega$ and V_Z remains a constant 12 V over the range of current values, for simplicity.

Do yourself

► FIGURE 3–14



Solution When $I_L = 0\text{ A}$ ($R_L = \infty$), I_Z is maximum and equal to the total circuit current I_T .

$$I_{Z(\max)} = I_T = \frac{V_{IN} - V_Z}{R} = \frac{24\text{ V} - 12\text{ V}}{470\ \Omega} = 25.5\text{ mA}$$

If R_L is removed from the circuit, the load current is 0 A . Since $I_{Z(\max)}$ is less than I_{ZM} , 0 A is an acceptable minimum value for I_L because the zener can handle all of the 25.5 mA .

$$I_{L(\min)} = 0\text{ A}$$

The maximum value of I_L occurs when I_Z is minimum ($I_Z = I_{ZK}$), so

$$I_{L(\max)} = I_T - I_{ZK} = 25.5\text{ mA} - 1\text{ mA} = 24.5\text{ mA}$$

The minimum value of R_L is

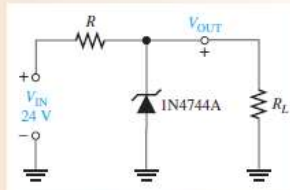
$$R_{L(\min)} = \frac{V_Z}{I_{L(\max)}} = \frac{12\text{ V}}{24.5\text{ mA}} = 490\ \Omega$$

Therefore, if R_L is less than $490\ \Omega$, R_L will draw more of the total current away from the zener and I_Z will be reduced below I_{ZK} . This will cause the zener to lose regulation. Regulation is maintained for any value of R_L between $490\ \Omega$ and infinity.

For the circuit in Figure 3–15:

- Determine V_{OUT} at I_{ZK} and at I_{ZM} .
- Calculate the value of R that should be used.
- Determine the minimum value of R_L that can be used.

► FIGURE 3–15



Solution The 1N4744A zener used in the regulator circuit of Figure 3–15 is a 15 V diode. The datasheet in Figure 3–7 gives the following information:
 $V_Z = 15\text{ V}$ @ $I_Z = 17\text{ mA}$, $I_{ZK} = 0.25\text{ mA}$, and $Z_Z = 14\ \Omega$.

(a) For I_{ZK} :

$$V_{OUT} = V_Z - \Delta I_Z Z_Z = 15\text{ V} - \Delta I_Z Z_Z = 15\text{ V} - (I_Z - I_{ZK})Z_Z \\ = 15\text{ V} - (16.75\text{ mA})(14\ \Omega) = 15\text{ V} - 0.235\text{ V} = \mathbf{14.76\text{ V}}$$

Calculate the zener maximum current. The maximum power dissipation is 1 W.

$$I_{ZM} = \frac{P_{D(\max)}}{V_Z} = \frac{1\text{ W}}{15\text{ V}} = 66.7\text{ mA}$$

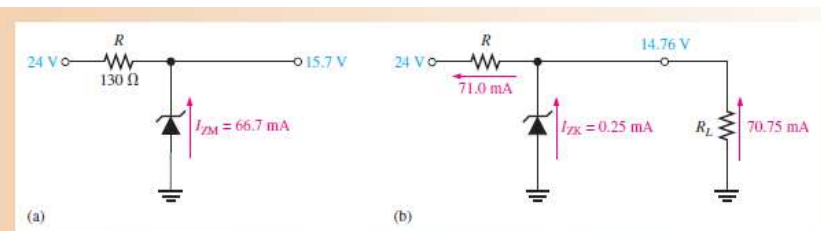
For I_{ZM} :

$$V_{OUT} = V_Z + \Delta I_Z Z_Z = 15\text{ V} + \Delta I_Z Z_Z \\ = 15\text{ V} + (I_{ZM} - I_Z)Z_Z = 15\text{ V} + (49.7\text{ mA})(14\ \Omega) = \mathbf{15.7\text{ V}}$$

(b) Calculate the value of R for the maximum zener current that occurs when there is no load as shown in Figure 3–16(a).

$$R = \frac{V_{IN} - V_{OUT}}{I_{ZK} \quad I_{ZM}} = \frac{24\text{ V} - 15.7\text{ V}}{66.7\text{ mA}} = 124\ \Omega$$

$R = 130\ \Omega$ (nearest larger standard value).



► FIGURE 3–16

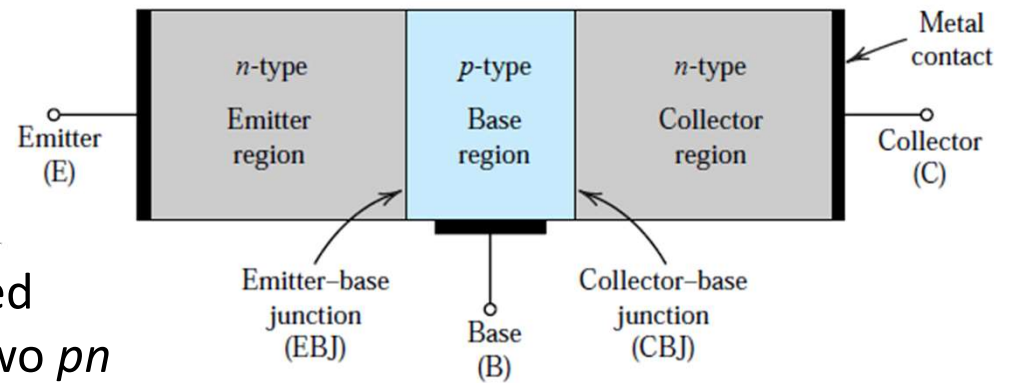
(c) For the minimum load resistance (maximum load current), the zener current is minimum ($I_{ZK} = 0.25\text{ mA}$) as shown in Figure 3–16(b).

$$I_T = \frac{V_{IN} - V_{OUT}}{R} = \frac{24\text{ V} - 14.76\text{ V}}{130\ \Omega} = 71.0\text{ mA}$$

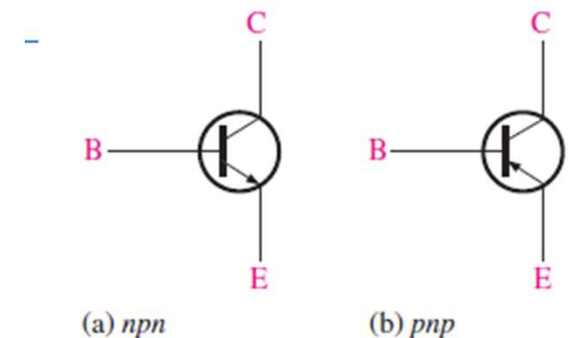
$$I_L = I_T - I_{ZK} = 71.0\text{ mA} - 0.25\text{ mA} = 70.75\text{ mA}$$

$$R_{L(\min)} = \frac{V_{OUT}}{I_L} = \frac{14.76\text{ V}}{70.75\text{ mA}} = \mathbf{209\ \Omega}$$

Bipolar Junction Transistors (BJTs)

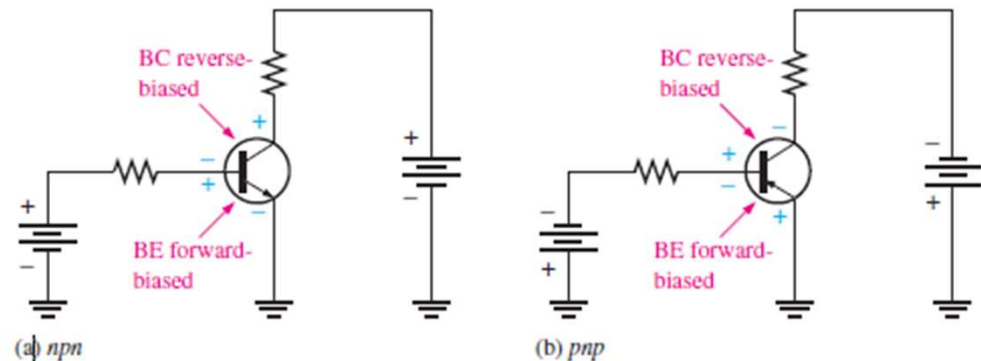


- The **BJT** is constructed with three doped semiconductor regions separated by two *pn* Junctions
- The three regions are called **emitter**, **base**, and **collector**.
- Why **bipolar** ?
- Why **Transistor** = transfer resistor ?
- The base region is lightly doped and very thin compared to the heavily doped emitter and the moderately doped collector regions.



Biasing

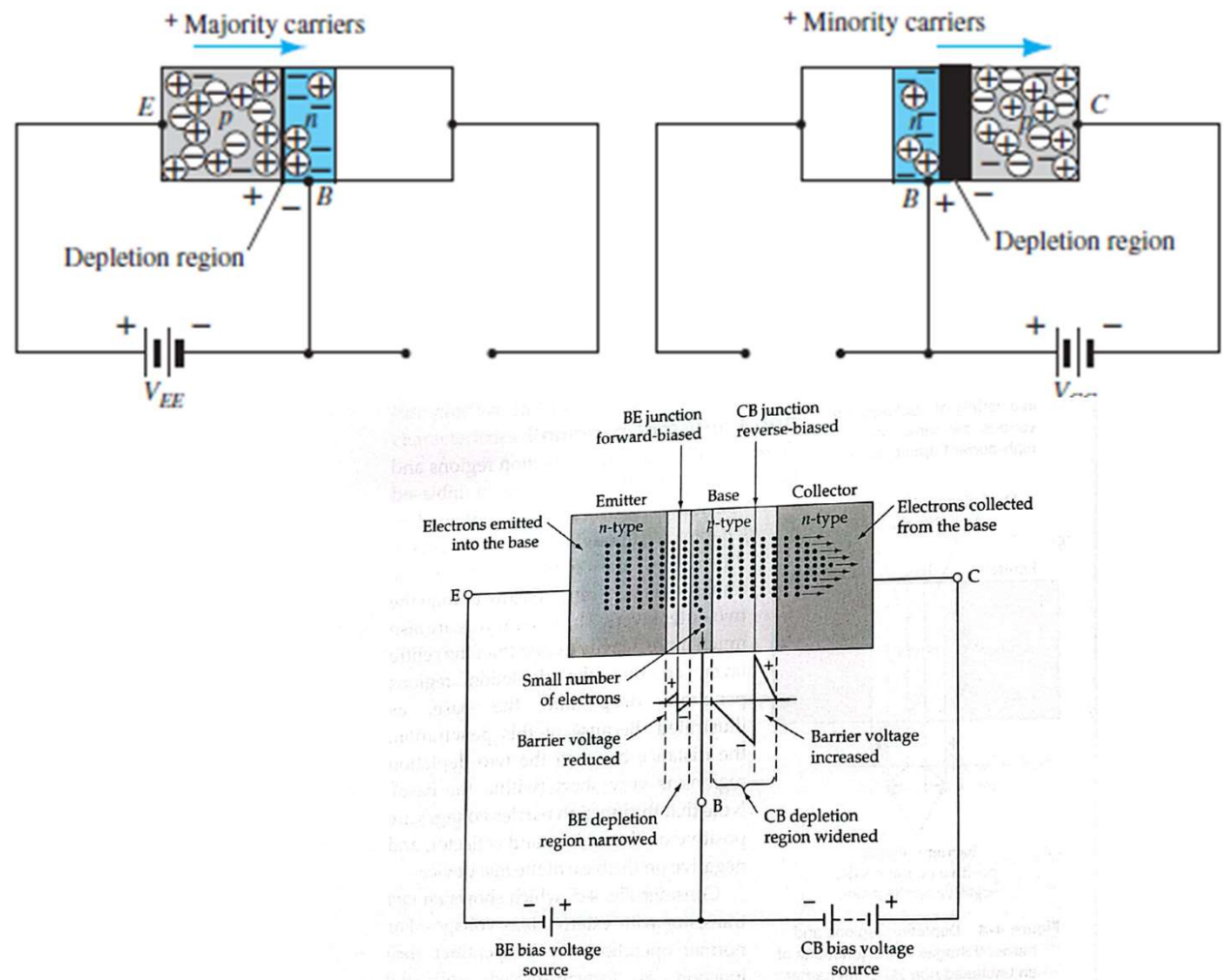
Mode	EBJ	ECJ
Cutoff	Reverse	Reverse
Active	Forward	Reverse
Saturation	Forward	Forward



Active mode(common emitter)

Operation

- Large number of majority carriers will diffuse across the forward biased p - n junction into the n -type material.
- Since the sandwiched n -type material is very thin and has a **low conductivity**, a very small number of these carriers will take this path of high resistance to the base terminal.
- The magnitude of the base current is typically on the order of microamperes, as compared to milliamperes for the emitter and collector currents.



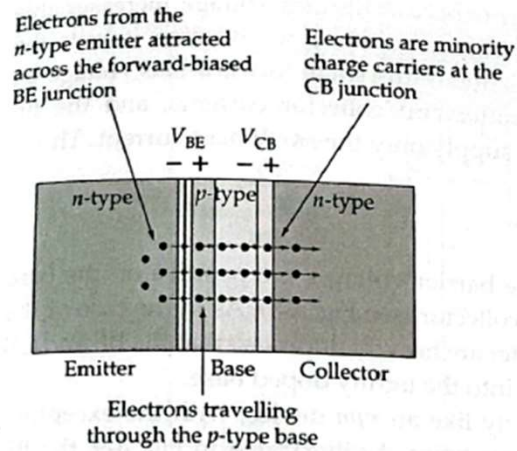


Figure 4-6 In an *nnp* BJT, electrons crossing from the emitter into the base region become minority charge carriers at the collector-base junction and are pulled across the collector-base barrier voltage.

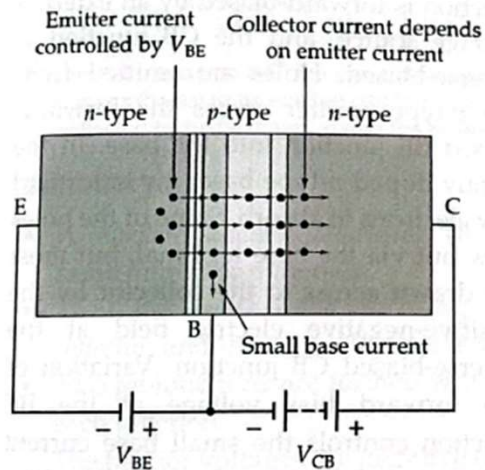


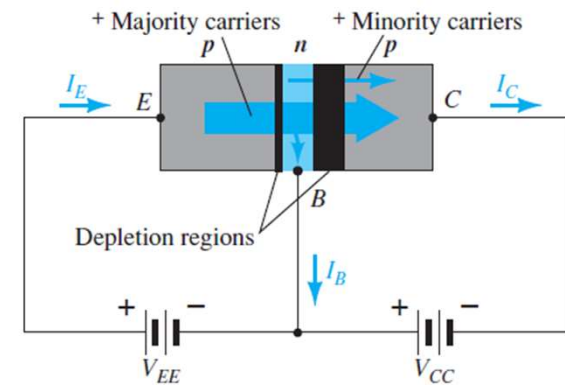
Figure 4-7 The quantity of charge carriers crossing from the emitter to the base is controlled by the base-emitter voltage. This means that the emitter and collector current levels are controlled by the base-emitter voltage.

- The larger number of these majority carriers will diffuse across the reverse-biased junction into the p -type material connected to the collector terminal.
- The injected majority carriers will appear as minority carriers in the n -type material.
- Applying Kirchhoff's current law to the transistor

$$I_E = I_C + I_B$$

The minority-current component is called the *leakage current* and is given the symbol I_{co} (I_C current with emitter terminal open).

$$I_C = I_{C_{\text{majority}}} + I_{CO_{\text{minority}}}$$



DC Beta (β_{DC}) and DC Alpha (α_{DC})

The ratio of the dc collector current (I_C) to the dc base current (I_B) is the dc **beta** (β_{DC}), which is the dc current **gain** of a transistor.

$$\beta_{DC} = \frac{I_C}{I_B}$$

Typical values of β_{DC} range from less than 20 to 200 or higher. β_{DC} is usually designated as an equivalent hybrid (h) parameter, h_{FE} , on transistor data sheets. h -parameters are covered in Chapter 6. All you need to know now is that

$$h_{FE} = \beta_{DC}$$

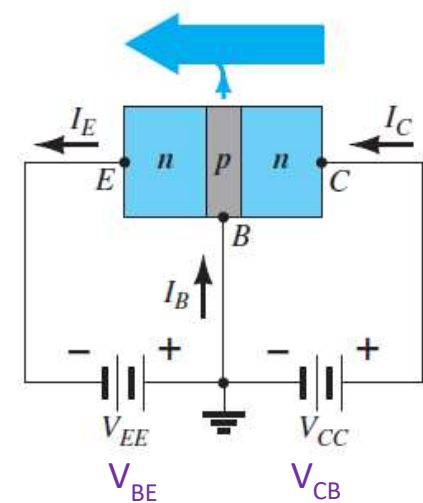
The ratio of the dc collector current (I_C) to the dc emitter current (I_E) is the dc **alpha** (α_{DC}). The alpha is a less-used parameter than beta in transistor circuits.

$$\alpha_{DC} = \frac{I_C}{I_E}$$

Typically, values of α_{DC} range from 0.95 to 0.99 or greater, but α_{DC} is always less than 1. The reason is that I_C is always slightly less than I_E by the amount of I_B . For example, if $I_E = 100$ mA and $I_B = 1$ mA, then $I_C = 99$ mA and $\alpha_{DC} = 0.99$.

COMMON-BASE CONFIGURATION

The common-base terminology is derived from the fact that the base is common to both the input and output sides of the configuration. In addition, the base is usually the terminal closest to, or at, ground potential.



To fully describe the behavior of a three-terminal device such as the common-base amplifiers of Fig. 3.6 requires two sets of characteristics—one for the *driving point* or *input* parameters and the other for the *output* side. The input set for the common-base amplifier as shown in Fig. 3.7 relates an input current (I_E) to an input voltage (V_{BE}) for various levels of output voltage (V_{CB}).

Input or driving characteristic

Levels of V_{CB} is making very small changes to V_{BE} .

So V_{BE} can be approximated to 0.7 once the transistor is “ON”.

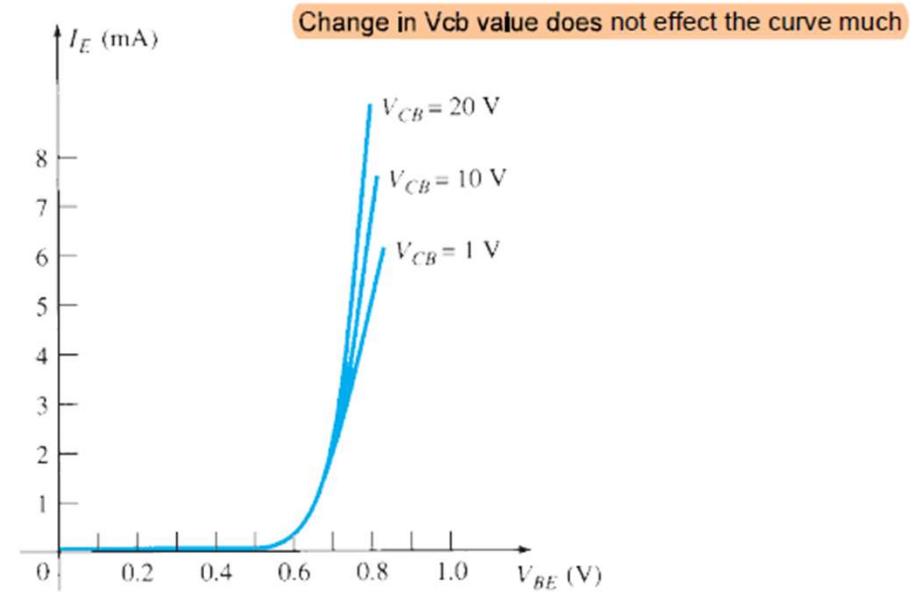


FIG. 3.7

Input or driving point characteristics for a common-base silicon transistor amplifier.

Output or collector characteristics

Three regions have been identified-

Active region

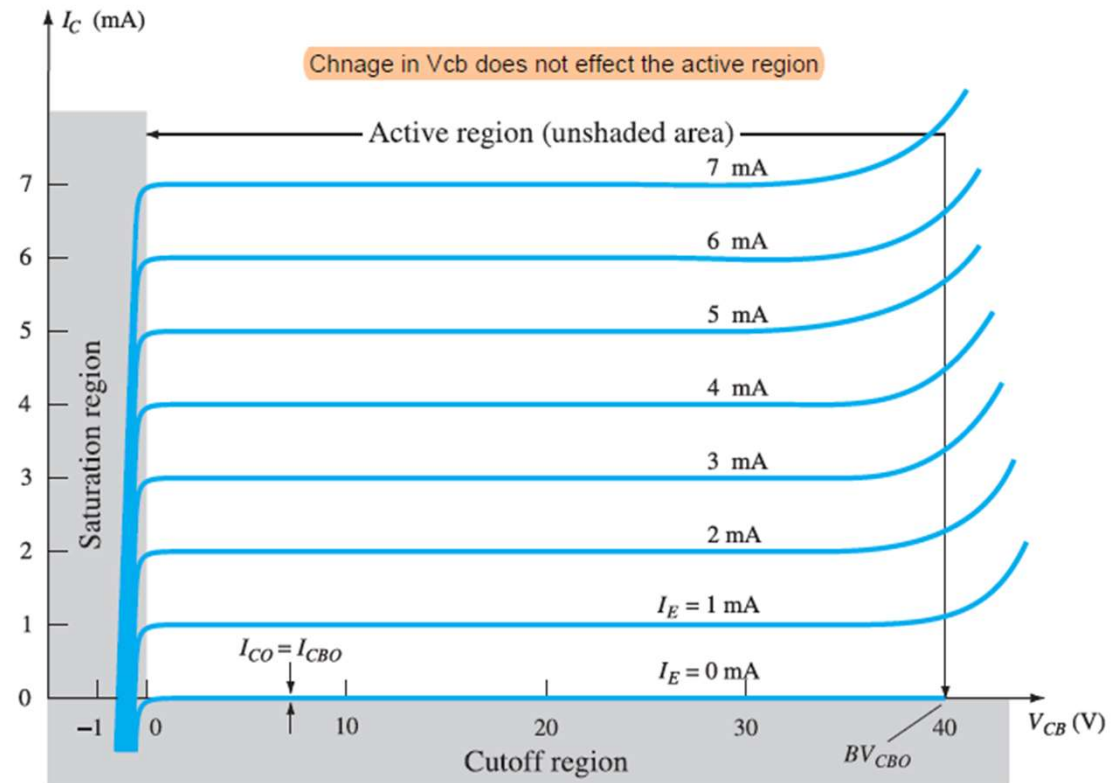
- Employed for the linear (undistorted amplifiers)
- $I_C \cong I_E$

Cut-off region

- $I_C = 0$

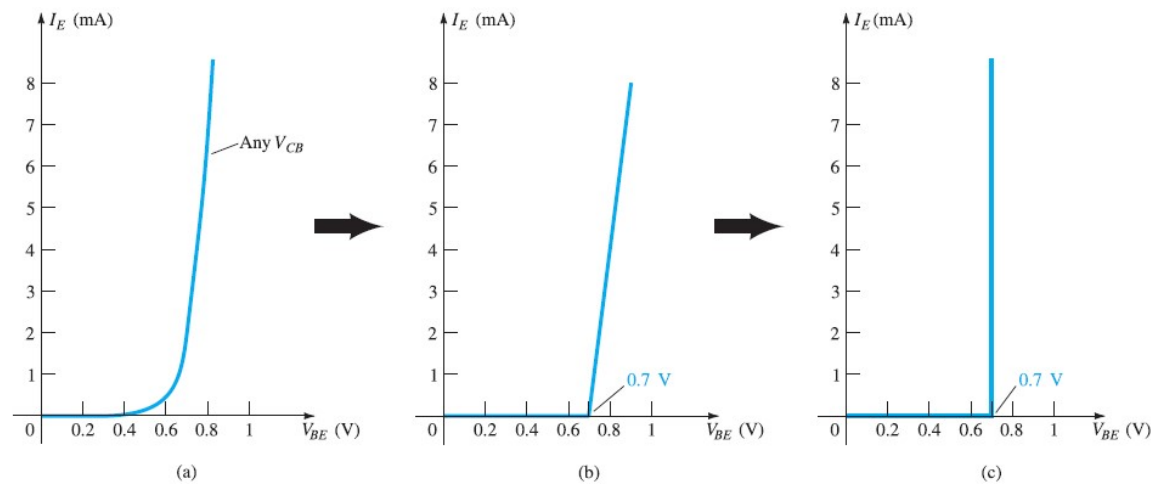
Saturation region

- Exponential rise in I_C as V_{CB} increases towards zero.



once a transistor is in the “on” state, the base-to-emitter voltage will be assumed to be the following:

$$V_{BE} \cong 0.7 \text{ V}$$



Book- Microelectronic Circuits by Adel S. Sedra and Kenneth C. Smith

Example 6.2

The transistor in the circuit of Fig. 6.14(a) has $\beta = 100$ and exhibits a v_{BE} of 0.7 V at $i_C = 1$ mA. Design the circuit so that a current of 2 mA flows through the collector and a voltage of +5 V appears at the collector.

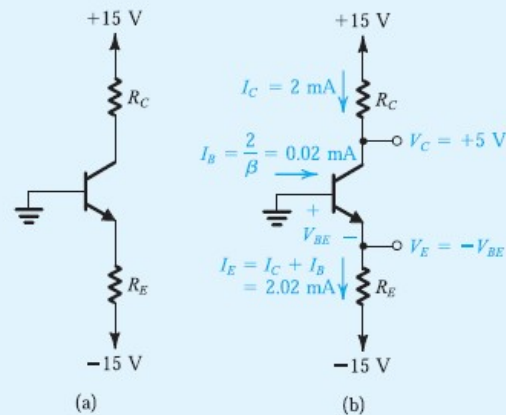


Figure 6.14 Circuit for Example 6.2.

Solution

Refer to Fig. 6.14(b). We note at the outset that since we are required to design for $V_C = +5$ V, the CBJT will be reverse biased and the BJT will be operating in the active mode. To obtain a voltage $V_C = +5$ V, the voltage drop across R_C must be $15 - 5 = 10$ V. Now, since $I_C = 2$ mA, the value of R_C should be selected according to

$$R_C = \frac{10 \text{ V}}{2 \text{ mA}} = 5 \text{ k}\Omega$$

$$i_C = I_S e^{v_{BE}/V_T}$$

in the Active Mode

Since $v_{BE} = 0.7$ V at $i_C = 1$ mA, the value of v_{BE} at $i_C = 2$ mA is

$$V_{BE} = 0.7 + V_T \ln\left(\frac{2}{1}\right) = 0.717 \text{ V}$$

Since the base is at 0 V, the emitter voltage should be

$$V_E = -0.717 \text{ V}$$

For $\beta = 100$, $\alpha = 100/101 = 0.99$. Thus the emitter current should be

$$I_E = \frac{I_C}{\alpha} = \frac{2}{0.99} = 2.02 \text{ mA}$$

Now the value required for R_E can be determined from

$$R_E = \frac{V_E - (-15)}{I_E}$$

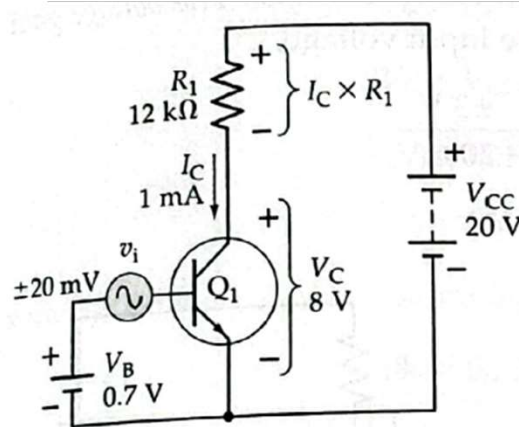
$$= \frac{-0.717 + 15}{2.02} = 7.07 \text{ k}\Omega$$

This completes the design. We should note, however, that the calculations above were made with a degree of precision that is usually neither necessary nor justified in practice in view, for instance, of the expected tolerances of component values. Nevertheless, we chose to do the design precisely in order to illustrate the various steps involved.

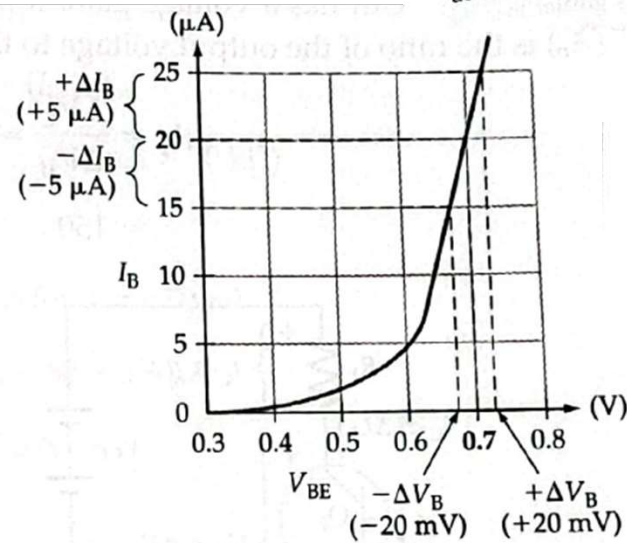
BJT Amplification

Voltage Amplification

Refer to the circuit in Fig. 4-17a, and assume that the transistor (Q_1) has $\beta_{dc} = 50$. Note that the 0.7 V dc voltage source (V_B) forward biases the transistor base-emitter junction. An ac signal source (v_i) in series with V_B provides a ± 20 mV input voltage. The transistor collector is connected to a 20 V dc voltage source (V_{CC}) via the 12 k Ω collector resistor (R_1).



(a) Transistor circuit with base bias and signal generator



(b) V_{BE} changes produce I_B changes

If Q_1 has the I_B/V_{BE} characteristic shown in Fig. 4-17b, the 0.7 V level of V_B produces a 20 μ A base current. This gives

$$I_C = \beta_{dc} I_B = 50 \times 20 \mu\text{A} = 1 \text{ mA}$$

The dc level of the transistor collector voltage can now be calculated as follows:

$$V_C = V_{CC} - (I_C R_1) = 20 \text{ V} - (1 \text{ mA} \times 12 \text{ k}\Omega) = 8 \text{ V}$$

While the ac input voltage (v_i) is zero, the transistor collector voltage remains at 8 V. When v_i causes a base voltage variation (ΔV_B) of ± 20 mV, the base current changes by ± 5 μ A, as shown in Fig. 4-17b. The I_B change produces a change in the collector current.

$$\begin{aligned}\Delta I_C &= \beta_{dc} \Delta I_B = 50 \times (\pm 5 \mu\text{A}) \\ &= \pm 250 \mu\text{A}\end{aligned}$$

Figure 4-18a shows that ΔI_C causes a change in the voltage drop across R_1 and thus produces a variation in the transistor collector voltage.

$$\begin{aligned}\Delta V_C &= \Delta I_C R_1 = \pm 250 \mu\text{A} \times 12 \text{ k}\Omega \\ &= \pm 3 \text{ V}\end{aligned}$$

The circuit ac input is the base voltage change (ΔV_B), and the ac output is the collector voltage change (ΔV_C). Because the output is greater than the input, the circuit has a voltage gain: it is a *voltage amplifier*. The *voltage gain* (A_v) is the ratio of the output voltage to the input voltage.

The circuit ac input is the base voltage change (ΔV_B), and the ac output is the collector voltage change (ΔV_C). Because the output is greater than the input, the circuit has a voltage gain: it is a *voltage amplifier*. The *voltage gain* (A_v) is the ratio of the output voltage to the input voltage.

$$\begin{aligned}A_v &= \frac{\Delta V_C}{\Delta V_B} = \frac{\pm 3 \text{ V}}{\pm 20 \text{ mV}} \\ &= 150\end{aligned}$$

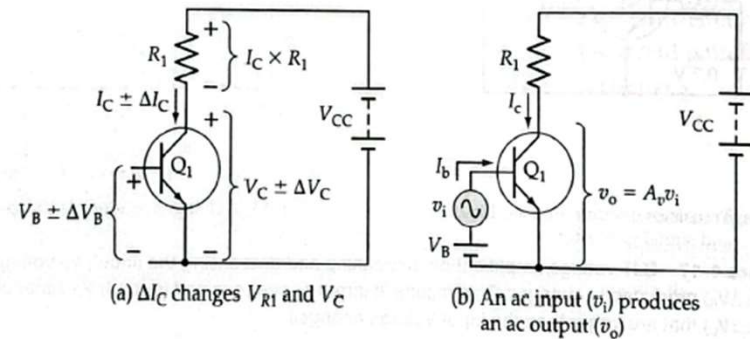


Figure 4-18 I_C changes produce changes in the voltage drop across the collector resistor (R_1), which result in V_C changes. This means that an ac input voltage (v_i) produces an amplifier output voltage ($v_o = A_v v_i$).

$$A_v = \frac{v_o}{v_i}$$

(4-8)

Substituting the ac quantities for Fig. 4-17a into Eq. 4-8 gives $A_v = 150$ once again.

Example 4-3

Determine the dc collector voltage for the circuit in Fig. 4-19a if the transistor has the I_B/V_{BE} characteristics shown in Fig. 4-19b, and $\beta_{dc} = \beta_{ac} = 80$. Calculate the circuit voltage gain when v_i is ± 50 mV.

Do yourself

Solution

From Fig. 4-19b,

$$I_B = 15 \mu\text{A for } V_B = 0.7 \text{ V}$$

Eq. 4-5:

$$I_C = \beta_{dc} I_B = 80 \times 15 \mu\text{A}$$

$$= 1.2 \text{ mA}$$

$$V_C = V_{CC} - (I_C R_1)$$

$$= 18 \text{ V} - (1.2 \text{ mA} \times 10 \text{ k}\Omega)$$

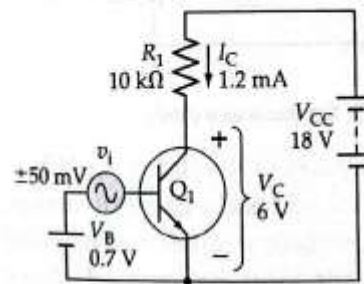
$$= 6 \text{ V}$$

From Fig. 4-19b,

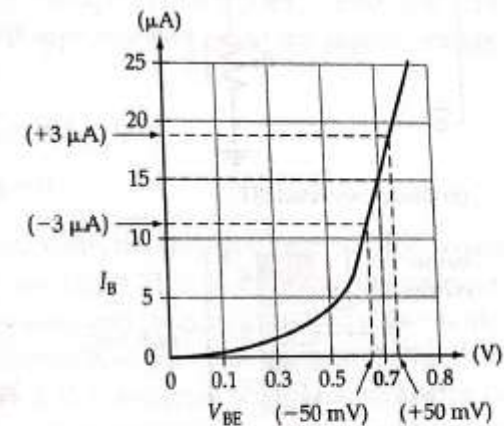
$$I_b = \pm 3 \mu\text{A for } v_i = \pm 50 \text{ mV}$$

$$I_c = \beta_{ac} I_b = 80 \times \pm 3 \mu\text{A}$$

$$= \pm 240 \mu\text{A}$$



(a) Transistor circuit



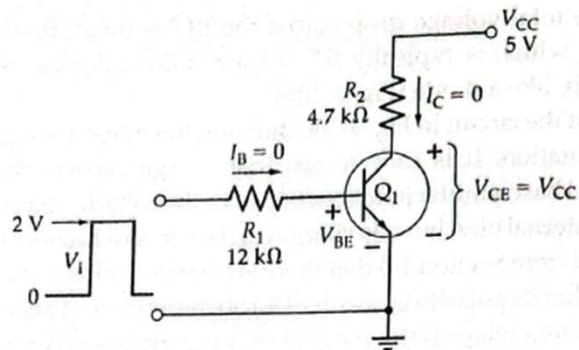
(b) V_{BE}/I_B characteristic for Q_1

Figure 4-19 Basic BJT amplifier circuit and I_B/V_{BE} characteristics for Ex. 4-3.

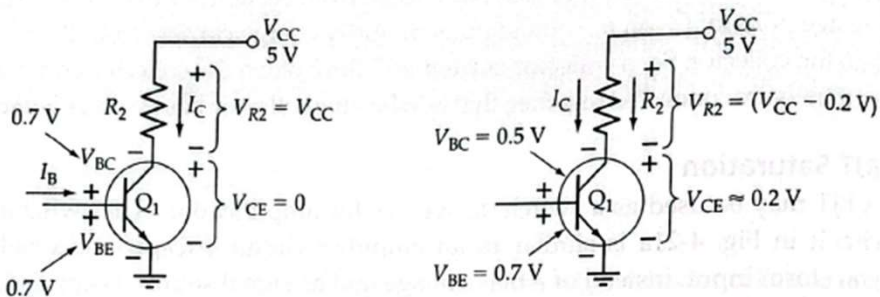
$$\begin{aligned} I_c R_1 &= \pm 240 \mu\text{A} \times 10 \text{ k}\Omega \\ &= \pm 2.4 \text{ V} \end{aligned}$$

Eq. 4-8:

$$\begin{aligned} A_v &= \frac{v_o}{v_i} = \frac{\pm 2.4 \text{ V}}{\pm 50 \text{ mV}} \\ &= 48 \end{aligned}$$



(a) BJT connected as a switch



(b) If $V_{R2} = V_{CC}$, $V_{CE} = 0$ and $V_{BC} = V_{BE}$

(c) When Q_1 is saturated $V_{CE} \approx 0.2$ V

Figure 4-21 When a BJT is employed as a switch, it is either biased *off* with V_{CE} equal to the supply voltage (V_{CC}), or *on* with V_{CE} equalling the collector-emitter saturation voltage ($V_{CE(sat)}$).

BJT switching

- Switching circuit resembles an amplifier circuit but employs a pulse waveform input instead of a bias voltage and AC signal.
- When input voltage (V_i) is 0, I_B is 0, resulting in $I_C = 0$ and $V_{CE} = V_{CC}$.
- $V_{CE} = V_{CC} - (I_C * R_2)$, with $I_C = 0$, $V_{CE} = V_{CC}$. (OFF state)
- Positive V_i leads to a base current flow.
- I_B is made large enough to produce an I_C level causing a voltage drop across R_2 to approximately equal V_{CC} .
- With $V_{CE} = 0$ V, the CB junction becomes forward-biased by 0.7V.(ON state)
- The circuit in Fig. 4-21a is a BJT switching circuit, where a small base current controls a much larger collector current to switch the transistor between off and on states.

Read yourself

The previous discussion about the diode-connected BJT shows that I_C flows when V_{CB} equals zero. It is found that even when the collector-base junction becomes partially forward biased, the barrier voltage is not completely overcome, and I_C continues to flow. A silicon *npn* transistor typically passes substantial collector current when $V_{BC} \approx +0.5$ V, which gives $V_{CE} \approx 0.2$ V (see Fig. 4-21c). In this situation, the collector-emitter voltage is termed the *saturation voltage* ($V_{CE(sat)}$), and $V_{CE(sat)}$ is the minimum level of collector-emitter voltage that exists when the voltage drop across the collector resistor (R_2) tends to drive V_{CE} toward zero.